

Project Submission Pro-Forma

Student name: Junyuan Yao

Student ID: 5736642

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- ☐ A glossary of terms (where appropriate)
- ☒ A clear statement of my project objectives
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Date: 2 September 2026

Design and Evaluation of a Complete-Outfit Recommendation System Using Frozen SigLIP Representations, a Lightweight Task-Specific Compatibility Head and Constrained Candidate Search

by

Junyuan Yao

Dissertation submitted in partial fulfilment for the Degree of Master of
Science in Applied Artificial Intelligence

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Declaration

I have read and understood the rules on cheating, plagiarism and appropriate referencing as outlined in my handbook and I declare that the work contained in this assignment is my own, unless otherwise acknowledged.

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Project definition for my degree: The MSc Applied Artificial Intelligence project must involve an investigation of a relevant AI topic that demonstrates technical depth, meets academic standards, and addresses a well-defined research question in one or more core AI areas—such as machine learning, deep learning, natural language processing, computer vision, generative AI, or AI for industry and robotics. It must also show critical awareness of the ethical, societal, or deployment implications of AI technologies. This can be achieved through the design, implementation, and evaluation of a technical AI solution or through a systematic and critical literature review that synthesises current research and practice. Regardless of the approach, students must apply appropriate research methods, engage critically with existing literature, demonstrate depth of understanding, and justify the relevance and rigour of their work in line with academic standards.

Relationship of this project to the degree definition: This project investigates a well-defined complete-outfit recommendation question through machine learning, deep visual representations and computer vision. It designs, implements and evaluates a system combining frozen vision–language features, a supervised compatibility model and constrained candidate search. Technical rigour is supported by controlled baselines, architectural ablations, repeated-seed testing and calibration analysis, while the literature review critically positions the approach against existing outfit-compatibility and recommendation research. The discussion examines data bias, domain shift, privacy, generalisation and deployment limits. The work therefore connects a technical AI solution with appropriate research methods, critical evaluation and responsible deployment considerations.

Abstract

Complete-outfit recommendation requires a system to model relationships among garments while respecting practical constraints and the item already selected by a user. This study evaluates whether a small project-specific compatibility head can support that task without retraining a large visual encoder or tying the output to a single retailer catalogue. Frozen 768-dimensional SigLIP image representations were combined with a 379,265-parameter compatibility head and trained on the published Polyvore Outfits disjoint compatibility split. Here, *lightweight* refers to the trainable head and its downstream training cost, not to the size or original pretraining cost of SigLIP. The model was compared with non-learned, linear, recurrent, type-aware and attention-based controlled baselines over three random seeds. Evaluation included classification, corrected FITB ranking, calibration, structural ablation, imbalance, overfitting, runtime, bias and domain shift.

The selected compatibility model achieved a held-out ROC-AUC of 0.8583 (95% bootstrap confidence interval 0.8524–0.8641) and 60.25% FITB accuracy (95% confidence interval 58.73–61.75). The three-seed mean FITB result was 59.94%. Removing the pairwise relationship module reduced mean FITB by 10.28 percentage points, with a paired 95% interval of 8.96–11.57 points; the smaller gain from slot embeddings was inconclusive. Training AUC was 0.9699. After validation-based early stopping, validation and test AUC differed by only 0.00145.

The model was integrated into an abstract, product-independent candidate space with weather and clothing-range constraints. Exact search exposed substantial loss from the previous quota path, which recovered the exact top result in 27.78% of 18 controlled queries. The findings support the feasibility of an auditable system with lightweight downstream training within the tested setting. Evaluation was limited by 29.50% official FITB coverage, sparse pure-menswear evidence, residual item-identifier overlap across the published split files, strong catalogue domain shift and the absence of participant evaluation.

Keywords: fashion recommendation; outfit compatibility; SigLIP; fill-in-the-blank; constrained search; responsible artificial intelligence

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Generative AI declaration

Generative AI tools were used in an assistive capacity during the development and preparation of this project. In response to task-specific instructions, they provided code solutions and debugging support, assisted with LaTeX formatting and diagram preparation, and supported language editing and refinement of selected dissertation text. AI outputs were not treated as academic evidence or used in place of cited literature or recorded experimental results. The student reviewed the retained outputs against the implemented code, saved experimental artefacts and cited sources, and remains responsible for the research decisions, analysis, interpretation, final wording and submitted work.

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Chapter 1: Introduction

1.1 Background and context

Digital fashion recommendation covers several related but distinct tasks, including the retrieval of visually similar products, the identification of substitutes, individual-item recommendation and the assessment of whether a set of garments forms a coherent outfit. Visual resemblance alone is insufficient for the final task. Two shirts may be substitutes because they serve a similar role, whereas a shirt, trousers and shoes may be complements because their different roles form a plausible ensemble. Early image-based recommendation research formalised this distinction between visual similarity, substitution and complementarity (McAuley et al., 2015). A modern review similarly characterises fashion recommendation as a domain in which visual, semantic, contextual and user-related signals interact, while identifying evaluation, explanation and bias as continuing challenges (Deldjoo et al., 2023).

Outfit compatibility research has moved beyond manually specified colour or category rules towards learned relationships among items. A bidirectional long short-term memory model represented outfits as ordered garment sequences and learned their compatibility (Han et al., 2017). Type-aware embedding spaces subsequently distinguished similarity from compatibility across different item relationships (Vasileva et al., 2018a). OutfitTransformer then used task-specific tokens and self-attention to model outfit-level relationships for compatibility prediction and complementary-item retrieval (Sarkar et al., 2023). These approaches demonstrate that complete-outfit recommendation is relational: a candidate should be assessed in the context of the other garments rather than scored as an isolated item. They also demonstrate that architecture, category representation, negative construction and evaluation task affect what a reported score means.

Large-scale vision–language pre-training provides a potential representation layer for smaller downstream studies. SigLIP uses independent sigmoid losses over image–text pairs instead of requiring softmax normalisation across a global batch, and was evaluated as a transferable vision–language representation by Zhai et al. (2023). A frozen SigLIP image encoder can supply reusable features while a smaller model learns the project-specific relationship among garments. Freezing does not make the representation neutral or guarantee transfer to fashion data; it instead reduces the number of trainable parameters and makes controlled comparisons feasible within the resources

of an MSc project.

Throughout this dissertation, *lightweight* describes the project-specific compatibility head and the cost of downstream training on precomputed features. It does not describe the size or original pretraining cost of the SigLIP encoder on which those features depend.

1.2 Problem definition and motivation

This dissertation studies a practical complete-outfit task. A user supplies one garment, and the system recommends abstract garment types and colours for the remaining supported positions. The prototype, named *Cove*, uses four logical slots: an inner top, an optional outer layer, a bottom and shoes. The selected garment remains fixed. Weather conditions, selected style and the user-selected clothing range constrain the candidate space, while alternatives allow the user to retain control over the final choice.

The learned problem is complete-outfit compatibility scoring followed by candidate ranking. This is separated from candidate feasibility. For example, weather logic may suppress an outer layer in hot conditions, and slot rules prevent two bottoms from occupying the same outfit. Such constraints should not be inferred indirectly from a compatibility score. The deployed system combines a learned relational score with explicit constraints and a deterministic fallback.

The candidate representation is also deliberately product-independent. Recommendations are expressed as abstract garment states defined by slot, broad type, colour, style and clothing range. The states are derived from secondary Polyvore training data. This design reflects the intended interaction: a recommendation can guide the use of an existing wardrobe or a later purchase without claiming that one retailer contains the only suitable solution. Product-independent candidate generation also creates a search problem because compatibility is a non-additive score over complete combinations. The study consequently examines whether bounded exact search produces materially different results from an earlier quota-limited retrieval path.

The motivation is both methodological and practical. A large end-to-end model would be difficult to train, compare and audit with the available time and computing resources. A rules-only system is transparent but cannot learn the relationships observed in outfits. The adopted middle position retains a frozen visual representation, trains a comparatively small compatibility head and evaluates the learned component separately from the contextual rules and search procedure.

Figure 1.1 summarises the system boundary from user and context inputs to the functions that produce and preserve recommendations. It also distinguishes learned compatibility ranking from recognition, explicit constraints and wardrobe management. Cove is treated as a context-aware complete-outfit recommender system. Search is the mechanism used to explore and rank feasible candidate combinations, rather than a separate product-search claim.

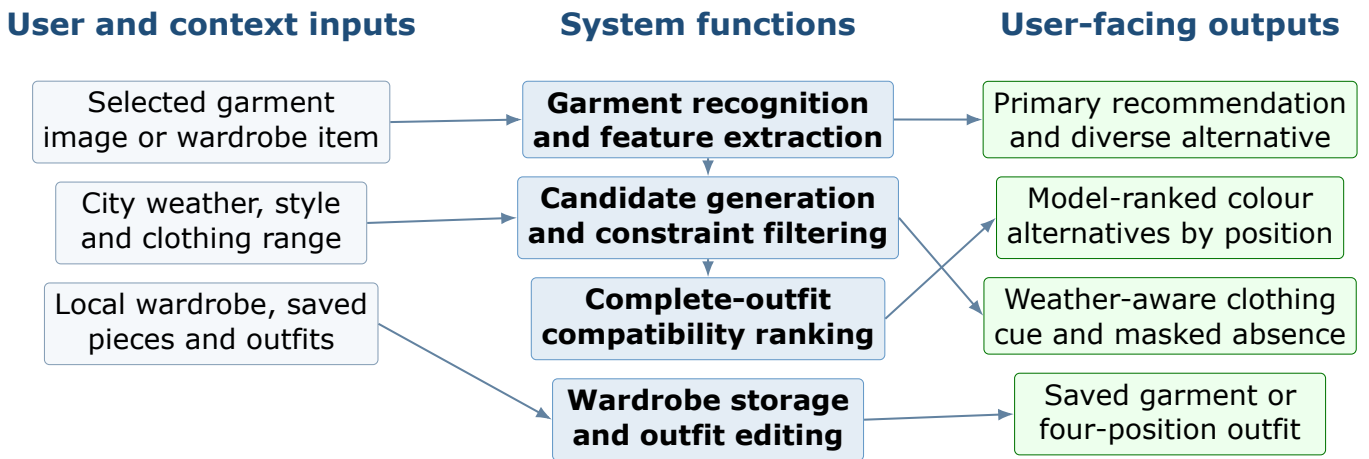


Figure 1.1: Functional scope of the Cove complete-outfit recommendation system

1.3 Research gap

The literature establishes that recurrent, type-aware and attention-based models can learn fashion compatibility (Han et al., 2017; Vasileva et al., 2018a; Sarkar et al., 2023). It does not follow that the most complex published architecture is automatically the most appropriate model for a smaller, processed dataset, a frozen visual representation and a resource-constrained application. Reported results are coupled to dataset splits, negative construction, candidate availability and evaluation protocol (Herlocker et al., 2004; Vasileva et al., 2018a). The broader fashion-recommender literature also cautions that offline accuracy alone does not resolve questions of explanation, bias or operational use (Deldjoo et al., 2023).

The gap addressed here is bounded and empirical rather than a claim that outfit recommendation itself is novel. First, it is uncertain whether a small relational head on frozen vision–language features can outperform non-learned and controlled learned alternatives under one consistent pipeline. Second, aggregate compatibility classification does not establish candidate-completion quality, calibration or generalisation; corrected FITB, ablation, repeated seeds and overfitting evidence are also required. Third, a model score does not by itself specify how a practical system should explore

a large abstract candidate space under weather and clothing-range constraints. The project addresses these questions jointly while keeping model evaluation distinct from claims about human taste or satisfaction.

1.4 Research question

The primary research question is:

To what extent can a small project-specific outfit-compatibility head built on frozen vision-language embeddings, combined with exact constrained search over abstract garment states, provide accurate, computationally practical and auditable complete-outfit recommendations using secondary fashion data?

Three sub-questions operationalise the model and system components:

RQ1.

How does the proposed compatibility model compare with non-learned and controlled learned baselines under consistent data and evaluation protocols?

RQ2.

Which architectural and training choices are supported by compatibility classification, corrected FITB, calibration, ablation and generalisation evidence?

RQ3.

What completeness, latency and control trade-offs arise when quota-limited retrieval is replaced by exact constrained search over product-independent abstract garment states?

The wording *to what extent* requires negative as well as positive evidence. A high AUC is treated as evidence of discrimination, not as the percentage of recommendations that users will like. Model ranking, uncertainty, runtime, data coverage, bias and domain shift all contribute to the answer.

1.5 Research objectives

The research question is addressed through the following objectives:

- 01.** Critically synthesise research on fashion compatibility, vision–language representation, recommendation evaluation and responsible design, and define a reproducible secondary-data method consistent with the approved ethical scope.
- 02.** Develop and evaluate a lightweight project-specific complete-outfit compatibility head using frozen SigLIP embeddings, including controlled baselines and evidence from classification, FITB, calibration, ablation, generalisation, imbalance and uncertainty analyses.
- 03.** Design and implement a product-independent recommendation prototype that combines exact candidate search, explicit contextual constraints, local wardrobe management, an interaction design intended to support clear and accessible use, and a deterministic fallback, then verify its functionality, coverage and runtime.
- 04.** Critically evaluate the confidence, limitations, practical value and future deployment opportunities supported by the combined model and system evidence.

1.6 Scope and boundaries

The investigation is confined to secondary datasets; no human participants were recruited for interviews, questionnaires or experiments. The project received confirmation dated 12 May 2026 that ethical approval was waived. The required research-integrity, information-security and WMG student-ethics training evidence, together with the waiver confirmation, is reproduced in Appendices A and B. The separately published Zara dataset retained locally is used for recognition and feature-domain diagnostics, not as the sole recommendation source.

The study does not claim to learn an objective or universal definition of fashion compatibility. Demographic attributes are not collected, so demographic fairness and user satisfaction cannot be measured (Mehrabi et al., 2021; Selbst et al., 2019). The interface asks the user to choose a menswear or womenswear candidate range; it does not infer gender from a person or garment image (Keyes, 2018). Sparse pure-menswear training data preclude a claim that both ranges have equal model reliability. Long-term personalisation from clicks is also outside scope because defensible online learning would require exposure logging, feedback-loop controls and a separate evaluation design (Chaney et al., 2018). Local wardrobe data do not retrain the compatibility model.

The evaluated representation is limited to an inner top, optional outer layer, bottom and shoes. Dresses, jumpsuits and other one-piece garments would require an explicit

multi-slot occupancy representation and separate training and evaluation. Accessories and culturally specific garment structures are not modelled comprehensively. Official FITB reporting is restricted to questions representable by these four slots and available frozen embeddings; coverage is reported so that the filtered result is not presented as a full official benchmark score.

The work evaluates offline compatibility and system behaviour, not commercial conversion or human preference. Earlier engineering models use different candidate or data protocols and are not treated as causal evidence that one development stage improves another. Controlled claims are based on same-split baselines, architecture comparisons, structural ablations and the exact-versus-quota search experiment.

1.7 Contributions

Within these boundaries, the dissertation makes four contributions. First, it provides a reproducible compatibility pipeline with lightweight downstream training that separates a frozen visual representation, a learned relational head and explicit contextual constraints. Second, it provides a controlled evaluation comprising non-learned and learned baselines, three seeds, corrected set-identifier-based FITB labels, structural ablation, calibration and an explicit overfitting audit. Third, it develops an abstract candidate representation and evaluates bounded exact search against a quota-limited retrieval path using the same deployed objective. Fourth, it demonstrates a local-first prototype in which the user retains the selected garment, chooses the clothing range, receives weather-aware alternatives and can fall back to deterministic rules if model artefacts are unavailable.

The contributions are methodological, empirical and artefactual. The comparison models are controlled adaptations rather than exact reproductions of the published architectures.

1.8 Dissertation structure

Chapter 2 critically reviews recommender-system foundations, fashion compatibility, vision-language representations, candidate search, evaluation and responsible-system concerns. Chapter 3 specifies the secondary-data pipeline, model, baselines, experimental protocol, metrics, candidate-search system and ethical safeguards. Chapter 4 reports the model and system results without extended interpretation.

Chapter 5 analyses model behaviour, architectural evidence, overfitting and search trade-offs. Chapter 6 relates the findings to previous research, answers the research questions and evaluates contribution, validity, practical implications and limitations. Chapter 7 concludes by assessing the objectives and identifying proportionate future work.

1.9 Summary

This chapter has framed complete-outfit recommendation as relational compatibility scoring followed by constrained candidate search. It has defined a bounded empirical gap, one primary research question, three sub-questions and four objectives. The scope excludes participant claims, online personalisation and unsupported garment structures, while retaining model, search, runtime, bias and reproducibility evidence. Chapter 2 now evaluates the literature from which the representation, compatibility and assessment choices are derived.

Chapter 2: Literature Review

2.1 Introduction

This chapter critically reviews the research underpinning the project’s representation, compatibility-modelling, retrieval and evaluation choices. Fashion recommendation includes visual similar-item retrieval, substitute recommendation, personalised ranking, outfit compatibility prediction and complementary-item retrieval (Deldjoo et al., 2023; Sarkar et al., 2023). Because these tasks differ in their inputs, supervision and definitions of success, evidence obtained for one cannot automatically support another.

The review first positions complete-outfit recommendation within recommender systems, then examines the development of fashion-compatibility models and the role of frozen vision–language representations. It subsequently considers learning objectives, evaluation, candidate search and responsible design. The literature is used to derive controlled alternatives, not to assume that the newest or most complex architecture is necessarily appropriate.

2.2 Positioning complete-outfit recommendation

Traditional recommender systems often learn from interactions such as ratings, clicks or purchases and ask which item should be presented to a particular user. Evaluation must be connected to the user task, available observations and intended use; otherwise, nominally similar experiments may not be comparable (Herlocker et al., 2004). Fashion recommenders extend this setting with visual, textual, contextual and social information (Deldjoo et al., 2023).

Cove has no longitudinal user–item history and therefore does not claim to learn stable individual preferences or compare collaborative-filtering algorithms. Its learned task is content-based outfit compatibility: estimating whether relationships within a set of garments resemble those observed in the training outfits. At runtime, that score ranks completions around a user-selected garment. Dataset compatibility and personal preference must remain distinct, because following a dataset regularity does not establish what one person will prefer.

Complete-outfit recommendation is consequently considered at three connected

levels. Representation converts garment images into features; relational scoring evaluates garments jointly; decision and interaction filter infeasible candidates, search combinations and expose alternatives. Secondary outfit data can evaluate the first two levels. The third can be tested for deterministic behaviour, completeness and runtime, but not for satisfaction without human evidence. This separation prevents the compatibility classifier from being treated as the entire recommender system.

2.3 Development of fashion-compatibility models

Early image-based recommendation distinguished visual similarity from complementarity. Items serving the same role may be substitutes, whereas items from different roles may be complementary despite being visually dissimilar (McAuley et al., 2015). Pairwise metric learning operationalises this distinction by learning spaces in which compatible pairs are close. Such models support efficient nearest-neighbour retrieval, but a complete outfit contains several interacting pairs. Summing local scores assumes that pairwise compatibility is sufficient to represent the set.

Han et al. (2017) represented outfits as ordered garment sequences and used a bi-directional long short-term memory network. This aggregates information from garments on both sides of an item, although outfits have no natural linguistic order and must be ordered through garment categories. The method therefore provides a useful structured baseline while retaining possible sensitivity to that imposed sequence.

Vasileva et al. (2018a) argued that compatibility depends on item type and proposed type-aware embedding spaces. A top-shoe relationship need not use the same transformation as a top-top relationship, making category or slot identity part of the learned relation rather than descriptive metadata. This supports testing slot-pair-specific projections and explicit slot representations.

Context-aware graph modelling provides another response to isolated pair relations. Cucurull et al. (2019) generated product embeddings conditioned on known compatible context and reported stronger compatibility prediction when more context was available. The finding supports the view that compatibility is conditional rather than intrinsic to one item. Graph construction requires available nodes and edges, which may not exist for a newly uploaded garment.

Attention allows each garment representation to interact with every other garment without requiring a fixed sequence. OutfitTransformer uses task-specific tokens and self-attention for outfit compatibility and complementary-item retrieval (Sarkar et al.,

2023). Treating an outfit as an unordered set is appropriate because rearranging the same garments should not change compatibility. Attention offers flexible higher-order interaction but also adds capacity. Results obtained under OutfitTransformer’s original protocol do not establish that equal capacity is required with frozen inputs and a smaller processed dataset.

These studies support several modelling assumptions rather than one universal architecture. Pairwise models prioritise local relations and efficiency; recurrent models learn structured aggregation; type-aware models encode category-dependent relations; and attention learns set-level interactions (Han et al., 2017; Vasileva et al., 2018a; Sarkar et al., 2023). Their differences motivate controlled comparisons using the same features, splits and training conditions.

2.4 Polyvore data and label construction

Polyvore datasets convert outfits assembled by platform users into observable item sets and are consequently practical secondary data for compatibility research (Han et al., 2017; Vasileva et al., 2018a). A positive example records an outfit retained in the dataset; it is not evidence of an objective or universally shared aesthetic judgement.

Negative outfits are commonly constructed by replacing an item in a positive outfit. Same-type replacement is harder than arbitrary-category replacement because the negative remains structurally plausible (Vasileva et al., 2018a). Since replacements are generated rather than independently judged incompatible, they may include false negatives. This risk follows from the published construction procedure; it does not establish that a known proportion of labels is incorrect.

Splitting the data changes the generalisation question. If a product appears in both training and testing outfits, a model may benefit from recognising an observed item. Vasileva et al. describe their revised Polyvore partition as a disjoint split intended to reduce this problem (Vasileva et al., 2018a). The present project retains that published name, while independently auditing the locally acquired metadata rather than assuming complete item-level separation. Such a split also leaves shared photographic conventions, category distributions, platform culture and labelling assumptions unchanged.

In the common fill-in-the-blank (FITB) task, one garment is removed and the model selects the correct completion from four candidates (Han et al., 2017; Vasileva et al., 2018a). FITB evaluates relative ranking within a small set and is closer to recommendation than binary classification, but remains smaller than full-catalogue retrieval. Out-

fitTransformer evaluates compatibility prediction and large-scale retrieval separately, reinforcing that FITB does not establish scalable retrieval quality (Sarkar et al., 2023).

Polyvore is therefore suitable for controlled training and offline evaluation. It cannot represent all clothing practices. Filtering official questions to four supported slots introduces an additional coverage boundary. The retained fraction must accompany the subset result so that it is not presented as the complete official benchmark.

2.5 Frozen vision–language representations

Earlier compatibility systems commonly used convolutional encoders pretrained on ImageNet and adapted to fashion data (Vasileva et al., 2018a). Vision–language models instead learn from large collections of image–text pairs. CLIP demonstrated transferable visual representations and zero-shot recognition across numerous downstream datasets (Radford et al., 2021).

SigLIP belongs to this family but replaces a global softmax contrastive objective with independent sigmoid losses over image–text pairs (Zhai et al., 2023). It was not pretrained for outfit compatibility; its image encoder instead supplies a general visual representation from which a smaller downstream model can attempt to learn garment relationships.

Freezing the encoder converts each image into a fixed vector and restricts training to the compatibility head. For this project, that reduces trainable parameters, repeated image-processing cost and representation variation across baselines. It also permits comparisons with identical inputs. These are computational and experimental advantages within the project, not evidence that frozen features are universally preferable.

The downstream head cannot recover information absent from the frozen vector. Transfer is not guaranteed by source-model scale alone: Kornblith et al. (2019) found that transfer depends on both representation and target task, with limited benefit on some fine-grained datasets. Web-scale pretraining can also transmit biases in the source data (Radford et al., 2021; Mehrabi et al., 2021). Frozen SigLIP is thus a resource-conscious hypothesis to test rather than an assumed fashion-specific solution.

2.6 Learning objectives and controlled baselines

Compatibility classification and recommendation ranking are related but distinct. Binary cross-entropy trains separation of labelled positive and negative outfits. Its score can order candidates, but the loss does not directly optimise order among competing completions. Pairwise ranking instead encourages a preferred observation to score above an alternative; Bayesian Personalised Ranking is one established formulation for implicit observations (Rendle et al., 2009).

Neither objective is automatically preferable here. Binary labels are directly available in the compatibility rows, whereas ranking requires defensible positive–negative pairs. If a synthetic negative is noisy, a ranking objective may optimise an unreliable comparison. The choice must be evaluated rather than inferred from the fact that runtime recommendation uses ranking.

The controlled baselines form a hierarchy. Chance provides a lower bound for four-choice FITB. Non-learned cosine similarity tests whether frozen features already contain useful structure. A linear model over pooled features tests whether learned weighting is sufficient without nonlinear interaction. These controls prevent the downstream architecture from receiving credit for information already present in SigLIP.

A recurrent baseline represents the sequence assumption of Han et al. (2017); a type-aware model represents relation-specific transformations from Vasileva et al. (2018a); and attention represents set-level interaction following Sarkar et al. (2023). These are controlled approximations, not exact reproductions, which would require the original encoders, preprocessing, objectives and hyperparameters.

Structural ablation addresses component contribution. Removing the pairwise module tests whether explicit item–item interaction adds evidence beyond pooled outfit information. Removing slot embeddings tests whether slot identity adds evidence beyond visual vectors. If removal does not produce a consistent decline across repeated runs, the component’s benefit remains uncertain.

2.7 Evaluation beyond a single score

Evaluation should match the task claimed (Herlocker et al., 2004). Compatibility classification, candidate completion, confidence interpretation and application behaviour therefore require separate evidence.

ROC–AUC measures discrimination across thresholds and can be interpreted as the

probability that a randomly selected positive receives a higher score than a randomly selected negative (Fawcett, 2006). It does not measure whether the correct completion ranks first in a candidate set, nor whether a score is a calibrated probability.

FITB evaluates completion more directly. In a balanced four-choice question, chance accuracy is 25%, but performance still depends on candidate and label construction. Correct answers must be matched through the documented dataset relationship rather than assumed to occupy a fixed array position. Retaining question-level outputs makes label correction and paired comparison auditable.

Calibration is separate from discrimination. Neural networks may rank examples correctly while remaining overconfident; temperature scaling was effective in several classification settings examined by Guo et al. (2017). Because a positive scalar temperature does not alter logit order, it can improve confidence interpretation without improving FITB ranking.

Stochastic training variation must also be reported. Parameter initialisation, data order and other benchmark stages can alter an otherwise unchanged experiment (Bouthillier et al., 2021). Cove uses three seeded runs for its principal architecture comparisons. This is a resource-constrained repeated-run design, not a complete characterisation of training variance.

Structural ablations are evaluated on the same FITB questions and matched seeds. Pairing preserves the full-ablation relationship on each evaluation unit instead of treating their predictions as independent (Dietterich, 1998). The project’s bootstrap averages the three seeded outcomes within each question and then resamples questions, following the bootstrap principle of estimating sampling uncertainty by resampling (Efron and Tibshirani, 1994). Its interval is conditional on those runs and does not represent every possible training seed.

Overfitting is examined through training, validation and held-out test behaviour. Validation-based early stopping is intended to stop iterative training when further training improvement is no longer accompanied by better validation performance (Prechelt, 1998). A small validation–test difference supports only the consistency of the selected checkpoint on the current test split, not general reliability. Stronger training than validation and test performance is consistent with training-set overfitting, although data composition can also contribute.

No offline metric establishes user benefit. Without participant evidence, claims remain limited to discrimination, controlled completion ranking, calibration, runtime, coverage and deterministic system behaviour.

2.8 Candidate search and explicit constraints

A model scores only candidates that reach it. If a strong candidate is removed by a frequency quota or similarity shortlist, later ranking cannot recover it. Candidate generation is therefore part of recommendation behaviour rather than neutral preprocessing.

Large-scale complementary-item systems commonly use embedding retrieval. Outfit-Transformer creates a target representation and applies nearest-neighbour retrieval (Sarkar et al., 2023). Vector indexes and approximate search make this efficient at very large scale, trading retrieval completeness for speed or memory (Johnson et al., 2021).

Cove instead assigns a non-additive score to a complete combination. An item that is not individually closest to the uploaded garment can still participate in the highest-scoring complete outfit, so independent slot shortlists may not preserve the final optimum. This follows from the deployed scoring function rather than from a claim made by the retrieval literature.

For a bounded four-slot space, batched exhaustive scoring can remain feasible. Here, *exact* means that every combination surviving the declared filters is evaluated under one deployed objective. A safety limit is required so an oversized search fails explicitly rather than presenting a partial prefix as complete.

Weather suitability, slot occupancy and the user-selected clothing range are feasibility requirements, not aesthetic preferences inferred from Polyvore. The model ranks feasible outfits, while deterministic rules define what is allowed. This keeps model behaviour, search completeness and functional constraints separately auditable.

2.9 Bias, personalisation and responsible design

Bias can enter through data selection, labels, pretrained representations, candidate construction and interaction design, and cannot be reduced to overall accuracy (Mehrabi et al., 2021). Clothing practice is also cultural, temporal and subjective (Deldjoo et al., 2023).

Polyvore reflects a historical platform population and may over-represent some categories, presentation styles and commercial photography. Synthetic same-type negatives add another assumption, while SigLIP carries uncertainty from web-scale pre-training. These sources justify an audit and explicit limitations, but do not permit

demographic fairness to be measured without demographic attributes.

Automatic gender inference would add an uncertain identity classification, a practice criticised in human–computer interaction research (Keyes, 2018). Cove asks the user to choose a menswear or womenswear candidate range instead. Those terms remain simplified catalogue filters and are not definitions of the user.

Local execution reduces external transmission of wardrobe data but does not resolve feedback bias. Exposure influences which clicks become observable, and simulation evidence shows that algorithmically confounded feedback can increase behavioural homogeneity without increasing utility (Chaney et al., 2018). Defensible long-term self-learning would need exposure logging, exploration, feedback-loop controls and separate evaluation, so it lies outside the frozen study.

Fairness cannot be detached from social context or treated solely as a property of an isolated component (Selbst et al., 2019). A proportionate response documents limitations, avoids identity inference, retains the selected garment, exposes alternatives and permits rejection or replacement.

2.10 Synthesis and research framework

The literature supports four conclusions. First, compatibility is relational: visual similarity is insufficient and category identity can form part of the relation, although maximum model complexity is not always justified (McAuley et al., 2015; Vasileva et al., 2018a; Sarkar et al., 2023). This supports evaluating a lightweight relational head against progressively stronger alternatives.

Second, frozen vision–language features offer a feasible representation strategy rather than a fashion-specific solution. They reduce training burden while retaining transfer, domain and inherited-bias risks (Radford et al., 2021; Zhai et al., 2023; Kornblith et al., 2019). The downstream head must consequently be compared with non-learned and learned controls.

Third, ROC–AUC, FITB, calibration, repeated runs, ablation, coverage and domain diagnostics answer different questions (Fawcett, 2006; Guo et al., 2017; Bouthillier et al., 2021).

Fourth, candidate generation and explicit constraints affect deployed behaviour as well as scoring. Approximate retrieval is appropriate at very large scale, but a bounded non-additive four-slot objective creates a testable case for exact constrained search (Sarkar et al., 2023; Johnson et al., 2021).

The remaining empirical questions are whether a small relational head on frozen SigLIP features can balance discrimination, completion ranking, runtime and auditability under one secondary-data protocol, and how much deployed-model quality is lost through quota candidate generation. Chapter 3 translates those questions into controlled experiments.

2.11 Summary

Cove has been positioned as a content-based compatibility and constrained-search system rather than a conventional personalised collaborative recommender. The literature shows a progression from pairwise visual relations to sequence, type-aware, contextual and attention-based modelling. It supports frozen vision-language features as a resource-conscious experimental layer while requiring explicit treatment of transfer, labels, evaluation and bias.

Chapter 3 applies this framework through the methodology used for model and system evaluation.

Chapter 3: Methodology

3.1 Research approach

This project used a quantitative secondary-data approach combined with software-system evaluation. The investigation examined whether a small project-specific head could learn outfit compatibility from published fashion data and whether its scores could support a practical recommendation application. The research therefore covered both model performance and the behaviour of the complete recommendation process.

The project did not conduct interviews, questionnaires or other participant research. The available evidence is limited to offline model evaluation, controlled comparisons and functional system testing.

The complete technical route is shown in Figure 3.1. It separates secondary-data preparation and frozen feature extraction from compatibility learning, controlled evaluation and application deployment.

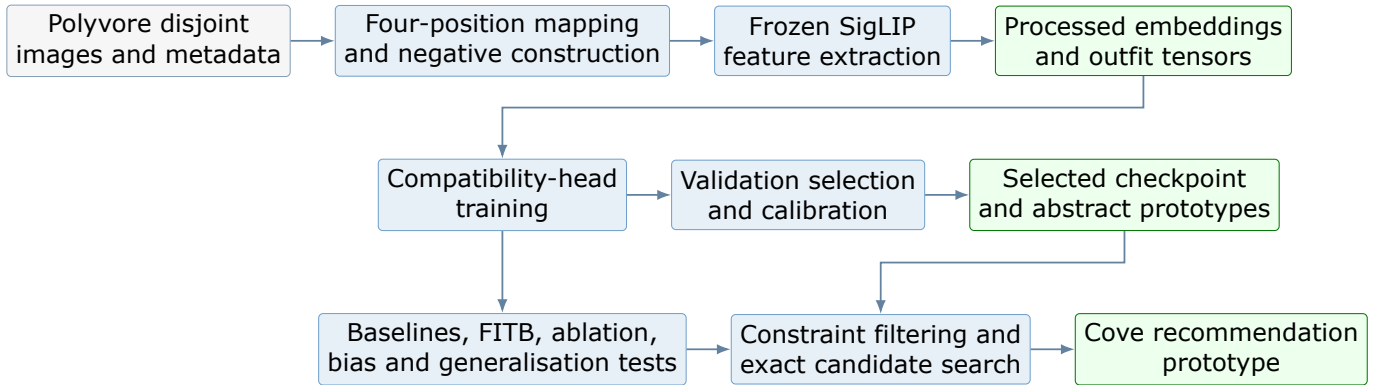


Figure 3.1: Technical route from secondary fashion data to the evaluated prototype

3.2 Data and ethical scope

The main model was trained and evaluated using files distributed as the Polyvore Outfits disjoint benchmark described by Vasileva et al. (2018a) and linked from the authors' repository (Vasileva et al., 2018b). The local copy was obtained on 21 August 2026. The three metadata files contain 16,995 training, 3,000 validation and 15,145 test outfits. Their outfit identifiers do not overlap. A reproducible local audit nevertheless found non-zero item-identifier overlap: 3,781 between training and validation,

84 between training and test, and 34 between validation and test. The dissertation therefore uses *disjoint* as the published split name and does not claim that the local artefact establishes strict product-level separation. This residual overlap is considered in the validity analysis.

The application supports four clothing positions: an inner top, an optional outer layer, a bottom and shoes. The published compatibility files contained 33,990 training, 6,000 validation and 30,290 test rows. Processing retained rows with at least three distinct supported positions and available embeddings, yielding 18,243 training, 3,274 validation and 16,062 test examples. The retained label counts were 9,121 positive and 9,122 negative training examples, 1,637 of each label for validation, and 8,031 of each label for test. The corresponding exclusions for insufficient supported positions were 15,747, 2,726 and 14,228 rows; 1,348, 220 and 1,414 of these rows also contained a duplicate supported position. No retained row lacked an embedding. Positive and same-type corrupted negative rows were read from the published compatibility file for their own split; negatives were not generated by drawing items across split boundaries.

Polyvore was used to learn compatibility relationships. The separately obtained *Zara Products* dataset was downloaded from its Kaggle record on 11 August 2026 (Parreño Lara, 2024). It was not added to the compatibility labels and was used only for garment-recognition preparation and feature-domain diagnostics. User-uploaded photographs remain on the local device and were not used as research data.

3.3 Recommendation-model development

SigLIP was used to convert each garment image into a numerical visual representation (Zhai et al., 2023). The implementation used `google/siglip-base-patch16-224`; the local Hugging Face cache resolved `main` to revision `7fd15f0689c79d79e38b1c2e2e2370a7bf2761ed` (Google, 2024). SigLIP itself was not retrained. Feature extraction ran without gradient updates, and the resulting 768-dimensional item vectors were cached before head training. The optimiser received only the project-specific head parameters, so encoder weights were never included in an update step. This substantially reduced the amount of computation required and allowed every comparison model to use the same visual information.

A small neural model was then trained to assign one compatibility score to a complete outfit. The model combines two forms of information: an overall summary of the garments and the visual differences between garment pairs. Clothing-position

information is also supplied so that, for example, the relationship between a top and shoes can be distinguished from the relationship between two other positions.

The compatibility head contains 379,265 trainable parameters. A 768-dimensional input is layer-normalised and projected to 256 dimensions with a linear layer and GELU. A learned 32-dimensional embedding for each of four clothing positions is concatenated to the projected item. The model forms a masked mean of the present items and a second masked mean of the six absolute pairwise difference vectors. Their concatenation is passed through a $576 \rightarrow 256 \rightarrow 128 \rightarrow 1$ multilayer perceptron, with GELU activations and dropout of 0.20 before the second linear layer. It was trained to distinguish positive outfits from constructed negative outfits. Its output is later used to rank recommendations. The training objective itself is compatibility classification rather than direct user-preference prediction.

Figure 3.2 shows the implemented compatibility head. The trainable component receives frozen item embeddings and a position mask, then combines a masked outfit summary with explicit pairwise differences before producing one compatibility logit.

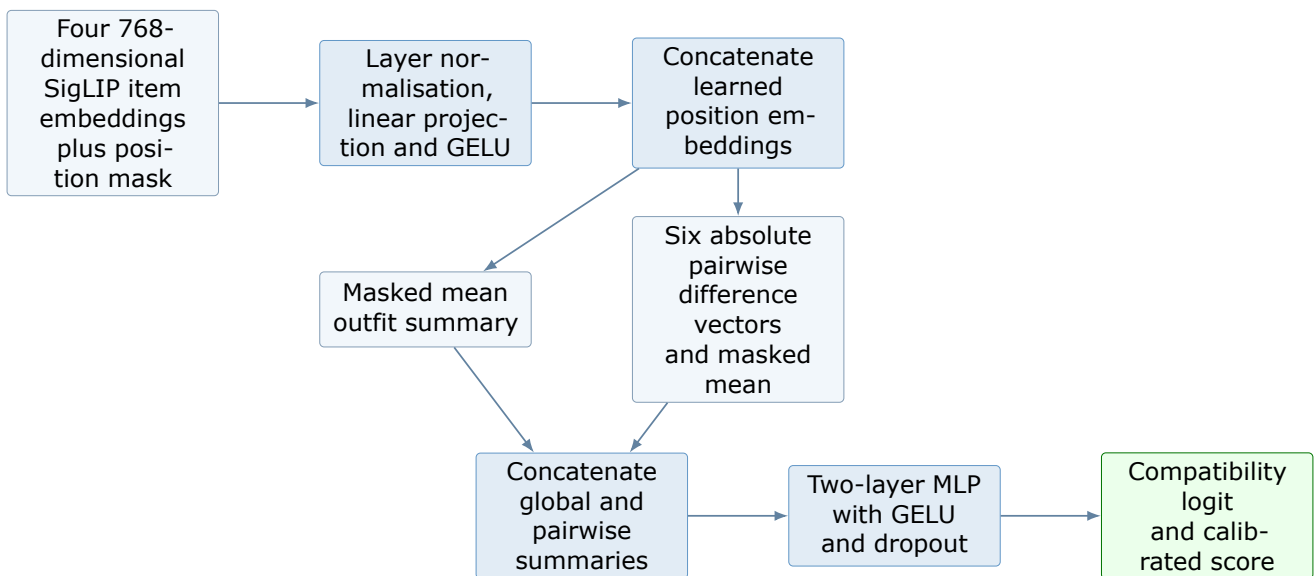


Figure 3.2: Architecture of the task-specific lightweight compatibility model

Training was limited to 12 epochs and used validation performance to select the saved checkpoint. Three predefined random seeds were used to examine whether results depended heavily on one training run. The test set was kept separate from model selection.

The processed labels were already balanced, so the final model did not apply additional positive-class weighting. Alternative weighting by outfit length was tested but did not satisfy the complete acceptance criteria. The unweighted model was therefore retained. Detailed settings and the rejected weighting experiment are reserved for

the appendices.

3.4 Comparative experiments

The proposed model was first compared with random ranking and a non-learned SigLIP similarity baseline. These controls test whether the model performs above chance and whether learning garment relationships adds value beyond the frozen visual representation.

Three further models represented established directions in outfit-recommendation research: a bidirectional LSTM based on Han et al. (2017), a type-aware pairwise model based on Vasileva et al. (2018a), and a Transformer-based set model informed by Sarkar et al. (2023). They used the same processed data, frozen image representations and training conditions as the proposed model. They are controlled implementations of the main architectural ideas rather than exact reproductions of the published systems.

Two ablation experiments were also conducted. One removed clothing-position information, and the other removed the module that explicitly compares garment pairs. The purpose was to determine whether these components made a measurable contribution when the rest of the experimental conditions remained unchanged.

3.5 Model and system evaluation

Compatibility classification was evaluated using several complementary measures. ROC-AUC measured how reliably positive outfits received higher scores than negative outfits (Fawcett, 2006). Accuracy, balanced accuracy, precision, recall, F1 and the confusion matrix were used to examine classification behaviour at the selected threshold. Calibration measures were reported separately because a model can rank examples correctly while still producing overconfident scores (Guo et al., 2017).

Recommendation performance was examined through the official fill-in-the-blank task. One garment is removed from an outfit and the model must select the correct replacement from four candidates. The correct answer was matched through the outfit identifier because the answers in the disjoint data are shuffled. Only questions compatible with the project’s four clothing positions were retained, and the retained proportion is reported with the result.

Confidence intervals were estimated by percentile bootstrap resampling (Efron and Tibshirani, 1994). The selected-model AUC interval used 1,000 resamples of the 16,062 test rows; the selected FITB interval used 2,000 resamples of 4,468 retained questions. Structural-ablation intervals used 5,000 paired resamples of those questions after averaging correctness over the three seeds. The exact-versus-quota score-loss interval used 2,000 resamples of the 18 query-level differences. All reported intervals use the 2.5th and 97.5th percentiles. The same questions and random seeds were used when comparing the complete model with an ablation, allowing their differences to be assessed directly. Training, validation and test behaviour were also compared to identify evidence of overfitting.

Expected calibration error used ten equal-width probability bins on the validation examples. Scalar temperature scaling minimised validation binary cross-entropy and did not use the test labels. Empty bins contributed no error; each populated bin was weighted by its share of validation examples.

The complete application was tested separately from the classifier. Candidate availability was checked under different weather and style conditions. Search experiments examined whether the previous quota-based method could miss higher-scoring combinations and measured the time required to search the complete filtered candidate space.

Figure 3.3 groups the evidence used to assess the model and the surrounding recommendation process. No single metric is treated as sufficient for deployment or for a claim about recommendation quality.

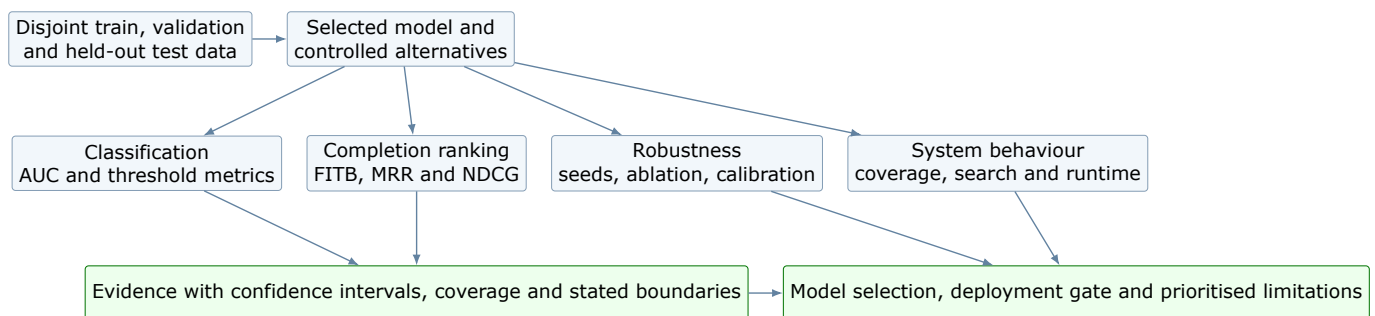


Figure 3.3: Evaluation framework connecting model, ranking and system evidence

3.6 Experimental controls and decision rules

The experiments were designed so that each comparison changed a defined part of the pipeline. Architecture comparisons used the same processed examples, frozen

SigLIP features, epoch limit, optimiser settings and predefined random seeds. This does not make the models identical in their learning behaviour, but it reduces the risk that a reported difference is caused by one model receiving different visual inputs or a more favourable data split. Parameter counts and recorded training times were reported to place performance in the context of model size and implementation cost. Training time was not treated as a hardware-independent benchmark.

Model selection was separated from final reporting. Validation AUC selected the saved checkpoint, and the test data were then used to estimate performance on unseen examples. The three-seed architecture experiment examined variation from initialisation and training order. Multiple runs cannot remove every source of benchmark uncertainty. They provide more reliable evidence than one favourable run (Bouthillier et al., 2021). Paired bootstrap comparisons reused the same resampled test examples for the two models being compared. This preserves the relationship between their predictions and is more informative than treating the scores as unrelated observations (Dietterich, 1998; Efron and Tibshirani, 1994).

Decisions about model changes used more than overall AUC. The length-weighting experiment had to retain overall discrimination, avoid reducing macro-level performance and improve the weakest length group. It was rejected when the weakest group deteriorated, despite a small improvement in overall mean AUC. The ranking-loss experiment was also not adopted because it did not improve the retained completion result. These decision rules reduce the temptation to select a configuration from one favourable number after seeing the results. They remain specific to the tested alternatives and do not imply that all weighting or ranking losses are unsuitable.

The search comparison also kept the scoring objective and candidate artefact fixed. Exact enumeration and quota-limited retrieval received the same held-out inputs and feasible candidate states. The exact path supplied the reference top results for the bounded space. Top-one recovery, top-ten recall, score loss and runtime then described what the quota path gained and lost. A higher model score was interpreted only as better optimisation of the deployed compatibility objective.

The 18 inputs were a deterministic coverage sample rather than a random or statistically representative sample. Positive examples in the processed disjoint test split were pooled by occupied clothing position in their stored order. The query index then cycled through all nine supported styles twice, rotated the fixed input across the four clothing positions, and cycled through the 12 interface colour labels. This produced five inner-top, five outer-layer, four bottom and four shoe inputs. The recorded run used the womenswear candidate range. Ground-truth garments from the held-out outfits were not inserted as candidate answers, and the script applied no result-based

filtering to the 18 generated cases. This sample supports a controlled engineering comparison of the two search procedures; its size and deterministic construction limit wider statistical generalisation.

3.7 Application integration, bias and reproducibility

The recommendation interface presents broad clothing concepts such as a black casual jacket or white trainers rather than claiming that the model has selected a particular commercial product. These concepts are derived from garments in the Polyvore training split and are displayed as coloured icons. Photographs are used only for garments uploaded or saved by the user.

Weather data were requested from the Open-Meteo forecast service (Open-Meteo, 2026). Weather, clothing position and the selected menswear or womenswear range are treated as explicit constraints. The application does not infer gender from a photograph. Because pure menswear training examples are limited, the same compatibility model is used for both ranges and no claim is made that their reliability is equal.

Reproducibility records include random seeds, model settings, training histories, software versions, file hashes and evaluation outputs. Automated tests cover data shapes, missing clothing positions, corrected fill-in-the-blank labels, clothing-range filtering, recommendation determinism and search safety limits. These records allow the reported experiments to be checked without relying only on the running user interface.

The software was organised so that training, evaluation and application inference could be run independently. Processed tensors and frozen embeddings were stored as explicit artefacts, rather than being recreated silently during each training run. Model checkpoints include the configuration required to reconstruct the compatibility head, while evaluation scripts write structured result files used by the tables in Chapter 4. This separation reduces the risk that interface state or later UI changes alter a reported experiment.

Integration used a guarded loading path. The application first checks whether the candidate artefact, model checkpoint and required metadata are present and compatible. If they are available, the learned score participates in ranking; otherwise the existing rule procedure is used and the fallback reason is recorded. The model was not allowed to replace category, weather or clothing-range validation. This design made it possible to test learned recommendation without making the whole application dependent on one model file. It also distinguishes reproducibility of an experimental checkpoint from availability of a user-facing feature.

The runtime sequence is shown in Figure 3.4. The selected garment remains fixed, contextual constraints define the feasible candidate space, and the learned model ranks complete combinations. A deterministic content-based path is retained only as an operational fallback when the required model artefacts cannot be loaded.

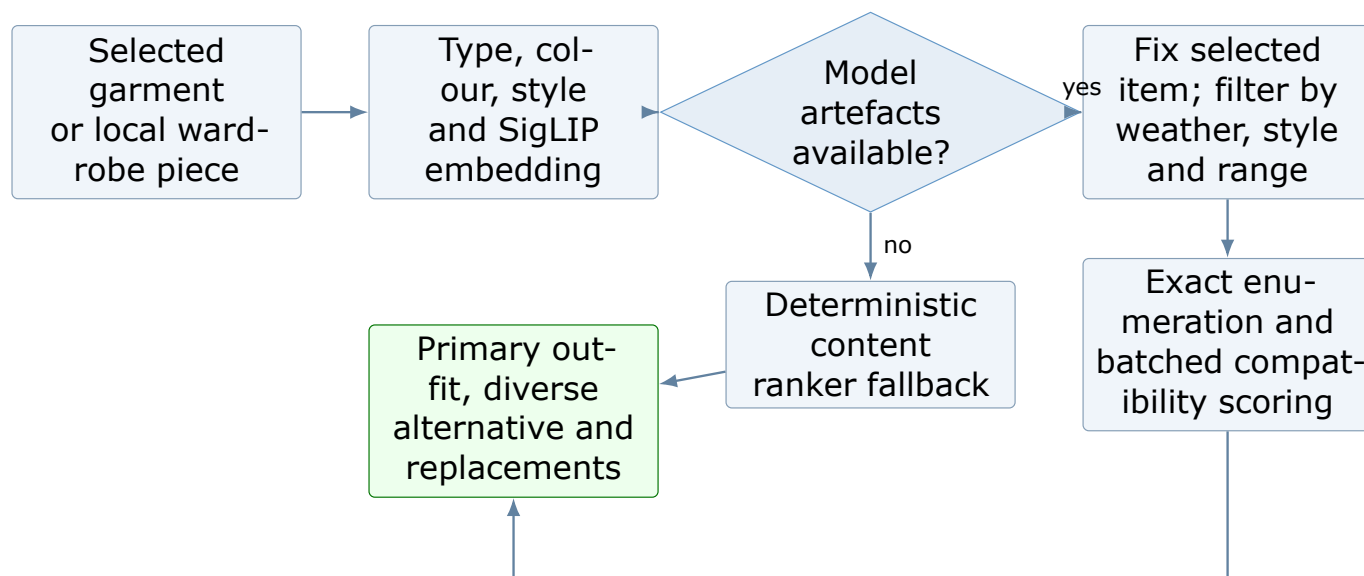


Figure 3.4: Runtime recommendation and fallback flow

The interface was designed to make the main workflow easy to follow without requiring knowledge of the underlying model. From the home page, the user can set or select a saved city, choose a visual mood, and continue either by starting with a garment or by opening the local wardrobe. In the recommendation path, the user supplies and selects the garment region, chooses a style and a menswear or womenswear candidate range, and then receives a ranked outfit with alternatives. Pieces and completed outfits can subsequently be saved and managed locally. Figure 3.5 shows the implemented entry point, including the weather scene, location controls, mood selector and two principal actions.

Visual hierarchy, large action controls and a short route to the two core tasks were used to support a convenient and visually coherent interaction. Five soft colour themes provide visual personalisation. The two-part clothing-range control also keeps menswear and womenswear recommendations under direct user control: the selected half is emphasised in the active theme colour, and the system does not infer gender from the uploaded image. These features were designed to support clear, accessible interaction. They are not reported as measured usability results because no participant evaluation was conducted.

Figure 3.6 shows the later stage of the same journey. The selected garment remains visible beside the controls, the chosen clothing range and style are explicit, and two

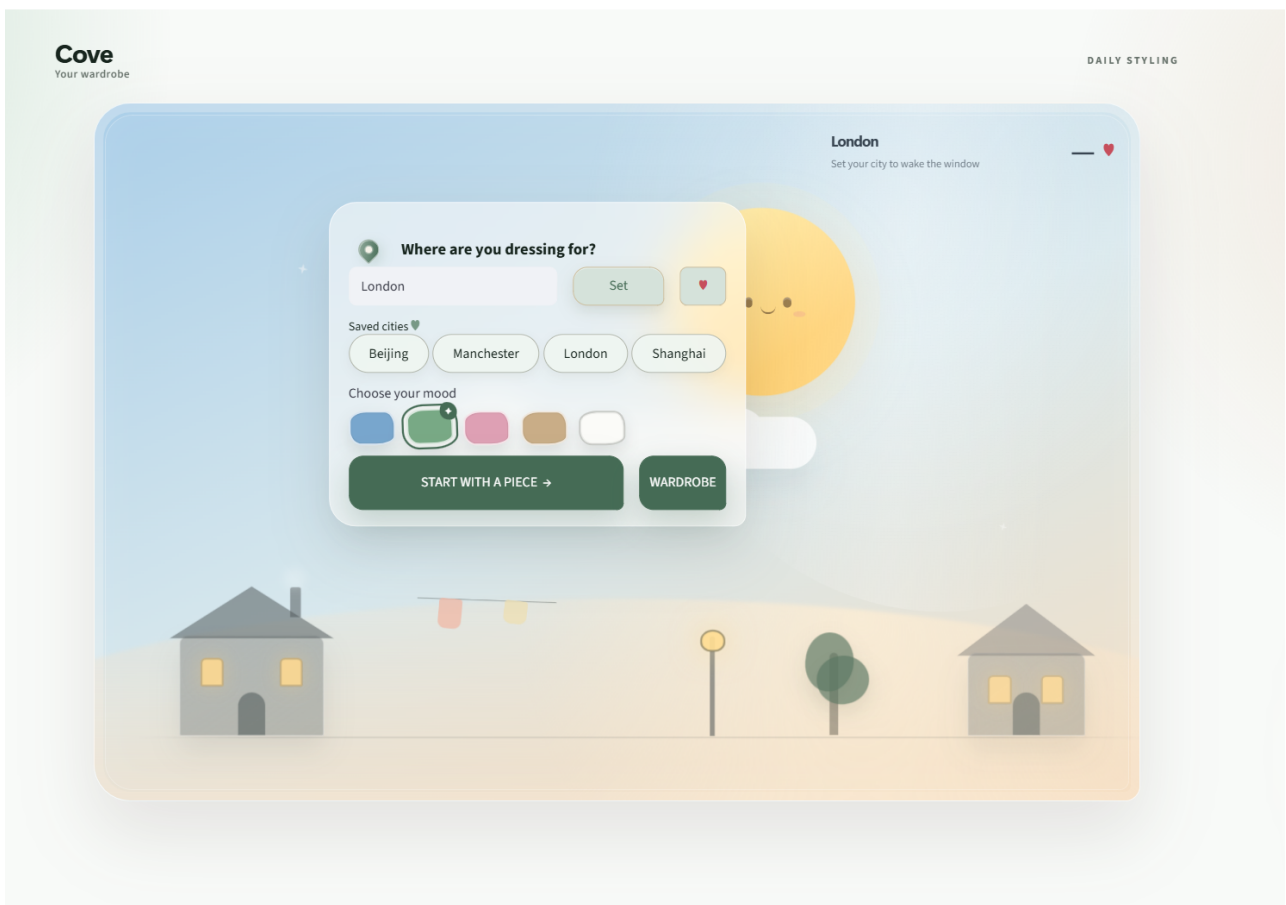


Figure 3.5: Cove home interface showing the weather scene, location controls, mood selector and routes to garment selection and the local wardrobe

complete outfit options are presented with abstract coloured garment icons. The user can request a model-ranked alternative for an individual position or save either complete outfit. An optional designer-discovery panel illustrates how a future product or independent-designer link could be added without changing the abstract recommendation itself.

Table 3.1 records the responsibility of each core application module. This division prevents the compatibility score from being described as if it also performs recognition, weather validation or storage.

Table 3.1: Core functions of the Cove prototype

Module	Main input and output	Responsibility
Garment analysis	Selected crop → type, colour, style and embedding	Interprets the item supplied by the user and prepares its model representation.
Candidate control	Context and fixed item → feasible states	Applies position, weather, style and clothing-range constraints.
Compatibility ranking	Complete candidates → ordered scores	Scores relational coherence using the small project-specific head.
Wardrobe management	Piece and outfit edits → local records and images	Saves, reuses, renames, groups and removes local wardrobe content.
Fallback	Recognised item and context → rule-ranked outfit	Preserves basic operation when learned artefacts are absent or incompatible.

Persistent state is implemented through JSON documents and local image files, while model and research outputs remain separate artefacts. Figure 3.7 presents these logical relationships without implying tables, foreign keys or a database engine that are not present in the implementation.

Supplementary software-design views are provided in Appendix E. They include the implemented use cases, overall component structure, recommendation-module dependencies, a logical entity-relationship view and traceable user stories. The diagrams describe the current prototype rather than an assumed enterprise architecture.

Result artefacts and checkpoints are larger than ordinary source files and may not always be suitable for repository distribution. File hashes and documented generation commands therefore provide an additional check that a locally held artefact corresponds to the evaluated version. This approach does not replace archival storage or full environment preservation, but it provides a practical chain from processed data to training history, evaluation output and deployed configuration.

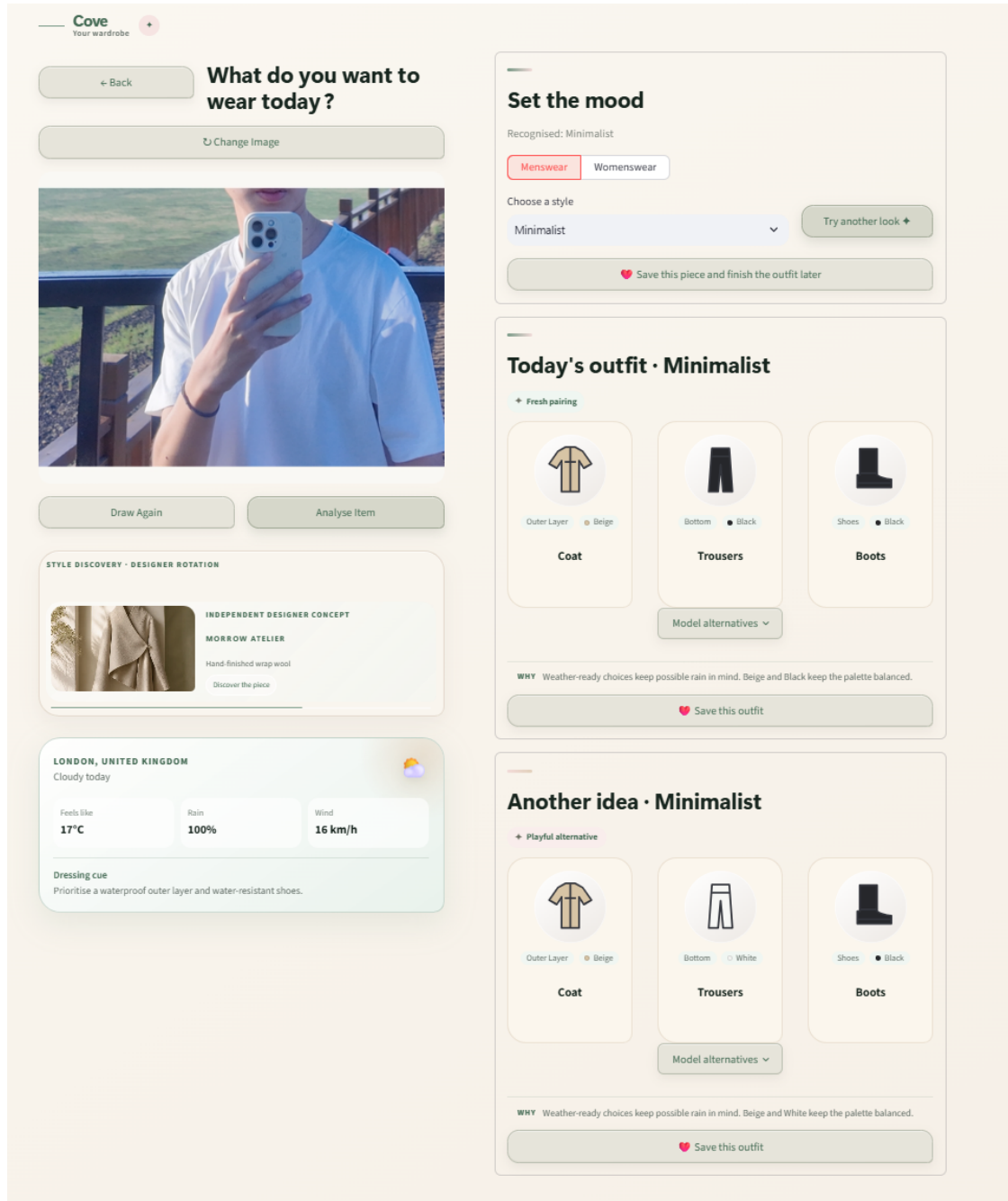


Figure 3.6: Cove recommendation interface showing the fixed garment, explicit clothing-range and style controls, abstract outfit alternatives and local save actions

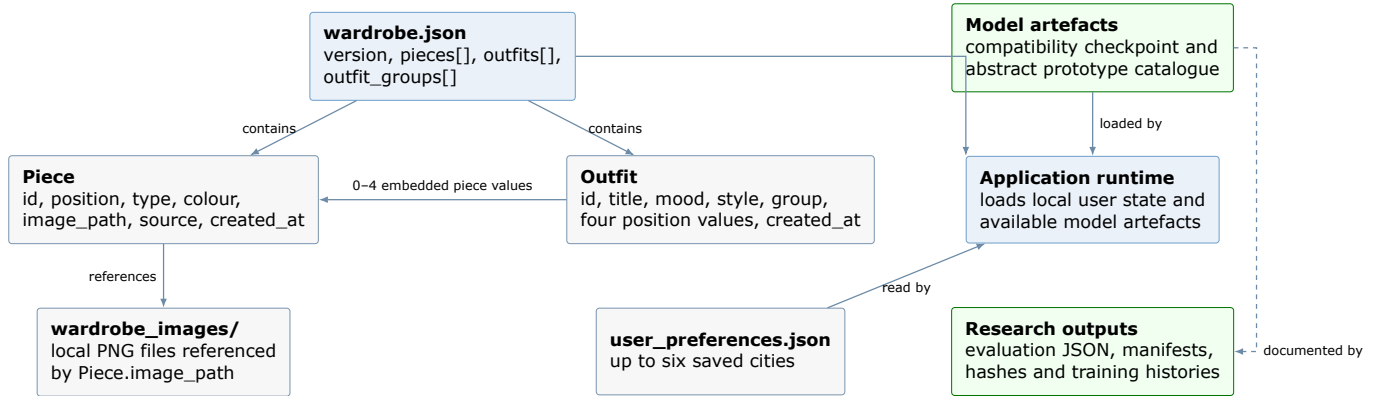


Figure 3.7: Logical relationships among local user data and model artefacts; the prototype uses files rather than a relational database

Table 3.2: Local persistence and research artefacts

File or artefact	Principal contents	Role and relationship
wardrobe.json	Pieces, outfits and outfit groups	Local user store; each outfit contains values for the four supported positions. Image paths are referenced by saved piece records.
wardrobe_images/	User garment PNG files	
user_preferences.json	Saved cities	Read by the application to restore local location choices.
Compatibility checkpoint	Model configuration, state and metadata	Reconstructs the selected compatibility scorer at runtime.
Abstract prototype artefact	Candidate states and frozen embeddings	Supplies the product-independent candidate space for constrained search.
Evaluation JSON and manifests	Metrics, histories, hashes and settings	Preserve the evidence chain; they are not user-profile records.

3.8 Research risks and mitigation

The investigation involved data, methodological, ethical and operational risks. These were considered during development rather than being listed only after the experiments. The first concerned data provenance and permitted use. The project therefore used documented secondary datasets and kept their roles separate. Polyvore supplied compatibility labels, whereas Zara was limited to recognition and feature-domain diagnostics. User photographs remained local and were not incorporated into the research dataset. Any later public release would require a separate check before third-party images or large derived artefacts could be redistributed.

Dataset availability was not treated as blanket permission to redistribute third-party content. The public-code package should exclude source dataset images, item-level embeddings and catalogue artefacts. The compact learned-head checkpoints contain model tensors, configuration and training metadata rather than source images or product identifiers; their redistribution should still be covered by a separate licence review before public release. This distinction keeps the ethical approval boundary, research use and publication permission from being treated as interchangeable.

A second risk was data leakage. Product overlap between training and test data could allow a model to recognise garments seen during training instead of learning relationships that transfer to unseen items. The published split was used to reduce this risk, and candidate concepts were derived from the training division rather than the test division. The local identifier audit confirmed zero outfit-ID overlap but found the residual item-ID overlaps reported above, so strict item-level separation cannot be claimed. Model selection and temperature scaling used validation data, while the test set was reserved for final evaluation. Repeatedly changing the model after inspecting test results could still weaken test independence. The current test results are therefore treated as final evidence for the selected model rather than as a further tuning target.

The labels create a further interpretive risk. Positive examples are observed Polyvore outfits, whereas negatives are constructed by replacing an item within a broad category. A constructed negative is not proof that an outfit is inherently unacceptable; it is a reproducible device for learning distinctions within this dataset. Similarly, official FITB evaluation was restricted to questions represented by the four supported positions and available embeddings. Coverage is reported with accuracy so that performance on the retained subset is not presented as a complete benchmark result. This follows the principle that recommender metrics must be interpreted in relation to the task and sampling protocol (Herlocker et al., 2004).

Overfitting and unstable model selection were addressed by recording training, validation and test behaviour separately, using three random seeds for the controlled architecture comparison and selecting checkpoints from validation AUC. Confidence intervals were estimated for the principal test and paired comparisons. Early stopping limits continued fitting after validation performance stops improving, but does not remove a generalisation gap (Prechelt, 1998). The results therefore report both the close validation–test result and the larger train–test difference.

Representation across clothing ranges was unequal. The retained FITB questions and pure training examples were concentrated in womenswear, so the small menswear subgroup could not support a reliable fairness comparison. The interface asks users to choose a clothing range and does not infer gender from an image. Menswear evidence is reported as sparse coverage rather than proof of equal reliability. This does not resolve representational bias, but it avoids automated gender classification and makes the evidential boundary explicit (Keyes, 2018; Mehrabi et al., 2021).

Finally, the deployed process could fail if model files, embeddings or candidate records were unavailable, or if constraints removed every candidate for a position. Release checks, deterministic seeds and recorded artefact paths support reproducibility. Slot, weather and clothing-range constraints reject infeasible combinations, while a deterministic rule-based fallback allows the application to continue when learned artefacts cannot be loaded. These mitigations make failure conditions more visible and bounded, without implying that technical testing removes all social or external-validity risks.

3.9 Summary

The methodology combines a frozen visual encoder, a small project-specific compatibility head and a constrained recommendation process. Controlled baselines, repeated training runs, fill-in-the-blank evaluation, ablation experiments and system tests were used to examine different aspects of performance. The following chapter reports the corresponding results.

Chapter 4: Results

4.1 Introduction

This chapter reports the compatibility, architecture, ablation, overfitting, bias and candidate-search results. Results are separated according to their experimental protocol. In particular, the selected compatibility model, the three-seed architecture comparison and the candidate-search experiment are reported independently because they answer different questions.

4.2 Selected compatibility-model performance

The selected compatibility model was evaluated on 16,062 held-out test examples containing equal numbers of positive and negative rows. It achieved a test ROC-AUC of 0.8583, with a 95% confidence interval from 0.8524 to 0.8641. Accuracy and balanced accuracy were both 0.7664. Table 4.1 reports the remaining classification results.

Table 4.1: Held-out performance of the selected compatibility model

Metric	Result
ROC-AUC	0.8583
95% AUC confidence interval	0.8524–0.8641
Average precision	0.8558
Accuracy	0.7664
Balanced accuracy	0.7664
Precision	0.7240
Recall	0.8610
F1	0.7866
Specificity	0.6718
Matthews correlation coefficient	0.5426
Brier score	0.1599

At the 0.5 threshold, the model correctly classified 5,395 negative and 6,915 positive outfits. It classified 2,636 negative outfits as positive and 1,116 positive outfits as negative. Recall was therefore higher than specificity at this threshold.

A separate fixed-seed baseline experiment produced an AUC of 0.4940 for determ-

inistic random scores and 0.7012 for mean pairwise SigLIP cosine similarity. The learned head in that experiment reached 0.8573. These values came from an earlier controlled baseline run and are reported separately from the selected model result of 0.8583. A mean-pooled frozen-SigLIP logistic baseline, evaluated on the same processed test rows, reached AUC 0.5532. Its deterministic preprocessing produced the same value in the three recorded seed files.

4.3 Controlled architecture comparison

Each learned architecture was trained with seeds 42, 7 and 123. The proposed model achieved the highest mean test AUC and the smallest AUC standard deviation across the three runs (Table 4.2).

Table 4.2: Controlled three-seed architecture comparison

Model	Parameters	Time (s)	Mean AUC	SD
Bidirectional LSTM	397,185	96.87	0.7928	0.0060
Type-aware pairwise	591,489	7.99	0.7918	0.0129
Set Transformer	374,145	7.03	0.7744	0.0133
Proposed model	379,265	4.81	0.8596	0.0009

The proposed model’s mean balanced accuracy was 0.7742 and its mean Matthews correlation coefficient was 0.5510. Mean balanced accuracy was 0.7138 for the bidirectional LSTM, 0.7176 for the type-aware model and 0.7034 for the set Transformer.

Paired bootstrap comparison of the three-seed mean predictions gave an AUC difference of 0.0700 between the proposed model and the bidirectional LSTM (95% interval 0.0650–0.0753). The difference from the type-aware model was 0.0634 (0.0578–0.0694), while the difference from the set Transformer was 0.0801 (0.0750–0.0858). These results compare controlled implementations on the project’s processed data rather than the headline results in the original publications.

4.4 Fill-in-the-blank and ablation results

Of 15,145 official disjoint FITB questions, 4,468 were compatible with the project’s four-position representation, giving coverage of 29.50%. Other questions were excluded because of unsupported answer positions, insufficient supported garments, duplicate positions or an answer position already occupied in the question.

The selected model achieved FITB accuracy of 0.6025. Its 95% bootstrap confidence interval was 0.5873–0.6175, compared with chance accuracy of 0.25. Mean reciprocal rank was 0.7705 and normalised discounted cumulative gain was 0.8289.

The structural ablation used the same 4,468 questions and three random seeds. The 0.6025 result above belongs to the selected seed-42 checkpoint; the complete model row below is the mean of three separately retrained seed runs and is therefore 0.5994. Table 4.3 reports the mean accuracy and the paired difference between the complete model and each reduced variant.

Table 4.3: Structural ablation on the retained official FITB subset

Configuration	Mean accuracy	Full minus variant	Paired 95% interval
Complete model	0.5994	—	—
Without clothing-position embedding	0.5947	0.0047	-0.0013–0.0106
Without pairwise module	0.4966	0.1028	0.0896–0.1157

The retained FITB data contained only 46 menswear questions, compared with 4,422 womenswear questions. Menswear performance was therefore recorded as a coverage limitation rather than treated as a reliable comparison between clothing ranges. The descriptive subgroup values are provided in the appendices.

4.5 Calibration and overfitting audit

The selected checkpoint reached its highest validation AUC at epoch 6. Training, validation and test AUC were 0.9699, 0.8598 and 0.8583 respectively (Table 4.4).

Table 4.4: Performance of the selected checkpoint by data split

Split	Examples	AUC	Accuracy	F1
Training	18,243	0.9699	0.8997	0.9051
Validation	3,274	0.8598	0.7688	0.7845
Test	16,062	0.8583	0.7664	0.7866

The train–test AUC difference was 0.1116, while the absolute validation–test difference was 0.00145. Validation AUC declined by 0.0100 between the saved epoch and the last observed epoch.

Validation-only temperature scaling produced a temperature of 1.5760. Validation negative log-likelihood decreased from 0.5081 to 0.4776, while expected calibration error decreased from 0.0772 to 0.0588. Candidate ordering was unchanged by the calibration step. Expected calibration error used ten equal-width probability bins over the 3,274 validation examples.

4.6 Outfit-length imbalance and feature-domain results

The class labels were balanced. A separate experiment examined the imbalance between three-item and four-item outfits. Table 4.5 reports the three configurations across the same three seeds.

Table 4.5: Outfit-length weighting experiment

Configuration	Overall AUC	Weakest-length AUC	Group gap
Unweighted	0.8596	0.8507	0.0393
Moderate length weighting	0.8601	0.8500	0.0433
Stronger length weighting	0.8553	0.8438	0.0485

Moderate weighting achieved the highest overall mean AUC but reduced the mean result for the weakest length group and increased the group gap. It therefore failed the predefined deployment gate, and the unweighted configuration was retained.

The feature-domain diagnostic compared 26,124 Polyvore test items with 300 Zara catalogue items. The domain classifier achieved cross-validated AUC of 1.000. Centroid cosine similarity was 0.7452, and the mean nearest-Polyvore similarity for the Zara items was 0.6528. The implications of this separability are considered in Chapter 5.

4.7 Candidate-search results

The product-independent candidate artefact contained 1,670 clothing concepts derived from the Polyvore training split, covering 23 garment types, 12 colours and nine styles. The selected womenswear range contained 1,508 concepts and the menswear range contained 162. Both ranges contained candidates for all four supported clothing positions.

The earlier retailer-derived candidate space contained 300 product records and 122 visible type-colour combinations. The product-independent representation contained

271 such combinations. Absolute compatibility scores across these candidate sources were not treated as directly comparable because their data distributions differ.

Exact and quota-limited search were compared on 18 held-out inputs using the same product-independent candidate space and compatibility model. The quota-limited path recovered the exact top result in 27.78% of queries and had mean recall of 24.44% for the exact top ten. It examined a mean 3.41% of the candidate combinations.

The mean score loss relative to the highest score found by exact search was 0.0179, with a 95% bootstrap interval of 0.0063–0.0357. Thirteen of the 18 queries had a positive score loss. Repeated exact searches returned the same top-ten results for every tested query.

Table 4.6: Exact-search and quota-limited search results

Measure	Quota-limited result
Exact top result recovered	27.78%
Mean recall of exact top ten	24.44%
Mean candidate space examined	3.41%
Mean score loss	0.0179
Queries with positive score loss	13 of 18

On the recorded CPU experiment, exact search had a median runtime of 0.474 seconds, a 95th-percentile runtime of 2.107 seconds and a maximum of 3.206 seconds. Quota-limited search over the same candidate space had a median runtime of 0.028 seconds.

Candidate availability was tested across four weather scenarios and nine styles, giving 36 combinations. Every tested combination retained all four logical clothing positions. In hot-weather cases, the outer-layer position was represented by the explicit masked *No Outer Layer* option.

4.8 Summary

The selected compatibility model achieved test AUC of 0.8583 and FITB accuracy of 0.6025 on the retained official subset. The three-seed comparison recorded a higher mean AUC for the proposed architecture than for the three controlled learned baselines. Removing the pairwise module produced a clear reduction, whereas the interval for removing clothing-position information crossed zero.

The selected model showed a substantial train–test gap but close validation and test performance. Validation-only calibration reduced negative log-likelihood and expected calibration error. The tested outfit-length weighting did not pass its deployment gate.

The product-independent candidate representation expanded the range of visible type–colour combinations. The quota-limited search recovered the same highest scoring result as exact search in 27.78% of the tested queries while requiring substantially less runtime. Chapter 5 examines the meaning and limitations of these results.

Chapter 5: Analysis

5.1 Introduction

This chapter analyses what can reasonably be concluded from the results in Chapter 4. It considers the evidence provided by the controlled baselines, structural ablations, held-out consistency audit and candidate-search experiment. It also distinguishes results that support the proposed design from those that reveal uncertainty or limitations.

The analysis remains within the scope of the experiments. A model score represents compatibility learned from the processed Polyvore data rather than an observed response from an individual user.

5.2 Compatibility performance and baseline evidence

The selected compatibility model achieved an AUC of 0.8583 on the held-out test data. This means that the model usually assigned a higher score to a randomly selected positive outfit than to a randomly selected negative outfit (Fawcett, 2006); it is not a percentage of correct recommendations.

The non-learned SigLIP similarity baseline reached an AUC of 0.7012. Frozen visual features contained some information related to the dataset labels without compatibility training. The learned model nevertheless performed substantially better under the same processed test protocol. This difference supports the need to learn relationships among garments rather than relying only on general visual similarity, consistent with the distinction between similarity and complementarity made by McAuley et al. (2015).

The proposed architecture also exceeded the three controlled learned baselines across the predefined seeds. Paired AUC intervals remained above zero for the comparisons with the bidirectional LSTM, type-aware model and set Transformer. Within this experiment, the result was not produced by one favourable initialisation or one unusually weak baseline run.

The comparison models were controlled adaptations using frozen SigLIP inputs and the project's four-position representation. The evidence shows that, under these shared

conditions, the small project-specific head was more effective than the selected alternatives. Direct comparison with published headline values would not be valid because the encoders, filtering and test coverage differ.

The proposed head was also the fastest of the four learned downstream architectures in the recorded training experiment. This supports the narrower claim of lightweight downstream training, subject to the recorded hardware, implementation and frozen-input pipeline.

5.3 Architectural evidence

Removing the pairwise module reduced mean FITB accuracy by 10.28 percentage points. The paired confidence interval remained entirely above zero. This provides the clearest structural evidence in the study: explicitly comparing garments contributed information that was not recovered by the overall outfit summary alone.

The finding accords with the relational treatment of compatibility in earlier work (Vasileva et al., 2018a; Sarkar et al., 2023). A garment may be plausible in one combination but not another, so an average outfit representation can lose differences between particular items. The pairwise component preserves a summary of these relationships without an attention-based interaction module.

Removing clothing-position information produced only a small reduction, and its confidence interval crossed zero. The experiment does not provide clear evidence that the position embedding improved FITB performance. The visual representation and pairwise relations may already distinguish broad garment roles, although the experiment did not isolate this mechanism.

The position embedding was retained because it adds little computational cost and gives the model an explicit indication of clothing role. The evidence does not establish it as an empirically necessary component. A simpler model without it remains a reasonable alternative for future evaluation.

5.4 Held-out consistency, overfitting and calibration

Training AUC of 0.9699 was considerably higher than held-out test AUC of 0.8583. The difference shows that the model fitted the training data more strongly than it performed on unseen test examples. The results contain evidence of training-set

overfitting.

Held-out test performance closely matched validation performance: their AUCs differed by only 0.00145, and the saved checkpoint corresponded to the highest observed validation AUC. This supports the consistency of validation-based checkpoint selection within the processed Polyvore split. Continuing beyond the selected epoch was not supported by the validation results. This is the limited role expected of early stopping: it controls continued fitting after validation performance peaks and cannot guarantee the absence of overfitting (Prechelt, 1998).

Further regularisation experiments should use validation-only decisions or a new untouched evaluation split, rather than tuning against the reported test results.

Temperature scaling reduced validation negative log-likelihood and expected calibration error. The original scores were more confident than necessary on the validation data. Calibration improves the interpretation of confidence but does not alter recommendation order or provide evidence of improved compatibility ranking (Guo et al., 2017).

5.5 Outfit-length imbalance

The dataset was balanced between positive and negative labels, so the principal imbalance did not concern class counts. Instead, three-item outfits were more common than four-item outfits. Moderate length weighting produced a slightly higher overall mean AUC but reduced performance for the weakest length group and increased the difference between the two groups.

Selecting that configuration only because its overall AUC was marginally higher would conflict with the purpose of the weighting experiment. The predefined gate required the weakest group to improve rather than allowing a small overall gain to hide weaker group performance. Retaining the unweighted configuration was consistent with the stated decision rule.

The tested weighting schemes did not satisfy the predefined balance gate.

5.6 Relationship between learned scores and explicit constraints

The application does not allow the compatibility score to decide every part of an outfit. This separation is necessary because statistical compatibility and feasibility answer

different questions. The model estimates whether a complete combination resembles compatible relationships in the processed training data. It does not know that an outer layer may be unnecessary in hot weather, that a user-selected garment must remain fixed, or that a candidate belongs to the requested clothing range. These conditions are applied before or during search rather than learned indirectly from the score.

The arrangement has two analytical advantages. First, a recommendation can be traced through candidate construction, constraint filtering and final scoring. If no outer layer is shown, the system can identify a weather mask rather than attribute the decision vaguely to the neural model. If an unsuitable clothing range appears, the error lies in filtering or metadata rather than in the compatibility score. This is a limited form of procedural explanation: it does not explain the visual features learned by SigLIP, but it identifies which system component made a deterministic decision.

Second, the rules stop the model from being evaluated for knowledge that its training objective did not provide. A positive or negative outfit label does not directly encode local weather, personal wardrobe ownership or application state. Expecting the compatibility head to infer these factors would mix separate sources of evidence and make failures harder to diagnose. The modular design supports auditability even though it may be less flexible than one end-to-end model.

This division also creates limitations. A rule can exclude a combination that the model would score highly, and errors in garment metadata can remove valid candidates before ranking. The reported model AUC does not test those rule failures. For this reason, model evaluation and 36-case candidate-availability testing were reported separately. The deterministic fallback is handled in the same way. It keeps the application usable when model artefacts are absent, but its output should not be combined with learned-model results or described as equivalent in quality.

5.7 Candidate-search completeness and latency

Quota-limited search examined an average of 3.41% of the feasible combinations. It recovered the same highest-scoring result as exact search in 27.78% of 18 queries and recovered an average of 24.44% of the exact top ten. The quota stage frequently removed combinations that the compatibility model would otherwise have ranked highly.

A positive score loss in 13 of the 18 queries confirms that the change was not limited to rearranging equally scored candidates. Score loss refers to the deployed model objective.

Exact search increased median runtime from approximately 0.028 seconds to 0.474 seconds on the recorded CPU experiment. Its 95th-percentile runtime remained close to two seconds. The experiment identifies a clear trade-off: exact search provides completeness under the model objective, while quota search provides lower latency. This reflects the broader retrieval trade-off between search coverage and computational cost (Johnson et al., 2021).

Within the tested candidate sizes, the additional runtime remained compatible with an interactive prototype. This statement applies only to the recorded hardware and bounded search space. A substantially larger catalogue could require approximate retrieval, staged ranking or another search strategy.

5.8 Coverage, domain shift and bias limitations

Only 29.50% of the official FITB questions could be represented by the four supported clothing positions. The resulting FITB accuracy is valid only for the retained subset and cannot be presented as performance on the complete official benchmark.

Menswear evidence was particularly sparse. Only 46 retained FITB questions were labelled as menswear, and the training data contained very few purely menswear outfits. The figures identify a coverage problem but do not estimate a dependable performance difference between clothing ranges.

The feature-domain classifier separated Polyvore and Zara items with an AUC of 1.000. Their frozen feature distributions were distinguishable under the diagnostic procedure. This does not measure how much compatibility accuracy would decline on Zara because compatible Zara outfit labels were unavailable. The result instead prevents an unsupported assumption that performance transfers unchanged between the two sources.

The candidate system avoids presenting historical product brands and allows the user to choose a clothing range, retain the uploaded garment and reject alternatives. These choices improve control and reduce some immediate product risks. Bias inherited from Polyvore labels or the frozen visual representation remains (Mehrabi et al., 2021; Selbst et al., 2019).

No participant evidence is available, so the findings are limited to offline model and system behaviour.

5.9 Summary

Overall, the pairwise module provided the clearest structural improvement, while evidence for the clothing-position embedding and weighting strategies remained inconclusive and the selected model retained an overfitting tendency. Exact search improved completeness under the deployed score at increased runtime; interpretation remains bounded by restricted FITB coverage, sparse menswear evidence, feature-domain separation and the absence of participant evaluation.

Chapter 6: Discussion

6.1 Introduction

This chapter brings together the literature, experimental findings and system evidence to answer the research questions. The discussion separates three levels of claim. The first concerns discrimination and completion ranking on the processed Polyvore data. The second concerns the contribution of model components under controlled ablation. The third concerns the behaviour of the complete prototype, including candidate search, constraints and fallback.

6.2 Relationship to previous compatibility research

Previous studies have represented an outfit as a sequence, as type-specific pair relationships, or as a set processed by attention (Han et al., 2017; Vasileva et al., 2018a; Sarkar et al., 2023). The present results support the premise shared by these approaches: compatibility is not adequately represented by isolated visual similarity. The non-learned cosine and mean-pooled baselines were weaker than the relational head, while the controlled three-seed comparison favoured the proposed head over the recurrent, type-aware and set-Transformer adaptations. This evidence supports the selected project-specific design under the shared protocol. The comparison models test architecture families under common tensors, frozen embeddings, labels and training conditions; they are not direct reproductions of published headline results produced with different experimental protocols. Under this project’s protocol, the small relational head was more effective and stable than the three controlled alternatives.

The result also depends on frozen SigLIP representations learned from large-scale image-text pretraining (Zhai et al., 2023). The project contribution therefore lies in the downstream relational modelling, controlled evaluation and system integration, while risks inherited from the pretrained representation remain relevant.

6.3 Answer to RQ1: comparative performance

RQ1 asked how the proposed model compared with non-learned and controlled learned baselines under consistent protocols. The evidence is positive within the defined experimental boundary. The selected checkpoint achieved test AUC of 0.8583, with a 95% confidence interval of 0.8524–0.8641. It was clearly above deterministic random scores and mean pairwise cosine similarity. Across three seeds, its mean AUC exceeded each controlled learned architecture; paired bootstrap intervals for the AUC differences did not include zero. Its standard deviation of 0.0009 was also lower than those of the comparison models.

These results indicate that the model learned a repeatable distinction between observed and corrupted outfits within the processed benchmark setting. AUC measures the probability that a randomly selected positive example is ranked above a randomly selected negative example (Fawcett, 2006). Same-category replacement reduces trivial category-based distinctions; compatibility remains defined by the available data and the published negative-generation protocol.

FITB accuracy of 0.6025 provides a second form of evidence because the model must select one answer from four alternatives. The result is above the chance level of 0.25, and its confidence interval does not approach chance. Coverage limits its interpretation because only 29.50% of the official questions were retained. The result applies to the four supported clothing positions with available embeddings, not to the complete Polyvore disjoint benchmark. Together, AUC and FITB demonstrate useful offline discrimination and completion ranking, while the coverage figure limits the scope of the latter result.

6.4 Answer to RQ2: architecture and training choices

RQ2 asked which design and training choices were supported by the combined evidence. The strongest structural result concerns explicit pairwise comparison. Removing the pairwise module reduced mean FITB accuracy by 10.28 percentage points, with a paired 95% interval from 8.96 to 11.57 points. The corresponding classification AUC also fell. This supports the view that compatibility depends on interactions among garments rather than only an outfit-level average.

The evidence for clothing-position embeddings was weaker. Removing them reduced mean FITB accuracy by 0.47 percentage points, and the paired interval ranged from –0.13 to 1.06 points. Since the interval crosses zero, the experiment does not establish

a reliable improvement. Its low cost and clear semantic role provide a practical reason for retention. The evidence is not strong enough to describe it as necessary. A larger or more position-diverse evaluation would be needed to resolve the effect.

The selected model was trained with binary cross-entropy on positive and negative outfits, then used as a scorer for candidate ranking. A scalar compatibility estimate can order candidates, so this use is coherent. It does not directly optimise FITB. Under the tested setup, the pairwise ranking-loss variant did not improve the retained evaluation and was not selected.

Held-out test performance closely matched validation performance, with an AUC difference of 0.00145. This supports the consistency of checkpoint selection within the processed split. Training AUC exceeded test AUC by 0.1116, and validation performance declined after epoch 6, showing that the model overfit the training distribution. Early stopping controlled the observed deterioration; it did not eliminate overfitting (Prechelt, 1998).

Temperature scaling reduced validation negative log-likelihood and expected calibration error without changing candidate ordering, consistent with its role as a post-hoc calibration method (Guo et al., 2017). These improvements apply to the validation distribution. Similarly, outfit-length weighting was rejected because a small gain in overall AUC coincided with weaker performance for the lowest-performing length group and a larger group gap. The decision follows the predefined acceptance gate rather than a change selected from overall AUC alone.

6.5 Answer to RQ3: candidate search and system control

RQ3 concerned the trade-off between quota-limited retrieval and exact constrained search over abstract garment states. In the 18-query comparison, the quota path examined only 3.41% of combinations on average. This reduced median CPU time from 0.474 seconds for exact search to 0.028 seconds. At this lower cost, it recovered the exact highest-scoring result in only 27.78% of queries and achieved 24.44% mean recall of the exact top ten. Mean score loss was positive, with a bootstrap interval that did not include zero.

The cases were generated deterministically to cover every supported style twice and every fixed clothing position at least four times. They were not a random sample of possible users or contexts. The comparison therefore provides evidence about the implemented search procedures on a bounded coverage set, not a population-level estimate of retrieval failure.

For the measured candidate space, exact enumeration improved completeness according to the deployed model objective, while retaining a median runtime below half a second and a maximum of 3.206 seconds. This supports its use in the current prototype. It does not show that exact search will remain practical for an unrestricted catalogue. The number of complete combinations grows multiplicatively with the candidates in each position, so later expansion may require pruning or approximate retrieval. Techniques for large-scale similarity search are well established (Johnson et al., 2021). Adopting one would introduce a new recall–latency trade-off that should be measured rather than assumed.

The abstract candidate representation also changed the role of recommendation. Polyvore did not become a catalogue from which users were instructed to select archived products. Instead, evidence from the training split was aggregated into product-independent combinations of clothing position, broad type, colour, style and clothing range. This matches the interface, which displays a coloured icon unless an item belongs to the user’s local wardrobe. Compared with the earlier retailer-derived space, the artefact increased visible type–colour combinations from 122 to 271. This shows broader representational coverage within the abstract representation.

Search remained subordinate to explicit feasibility rules. The user’s selected item was fixed, weather logic could mask the outer-layer position, and clothing range was selected manually. All 36 weather–style coverage cases retained the four logical positions. These checks support functional completeness within the defined representation.

6.6 Achievement of the research objectives

O1 was met through a critical review that informed a reproducible secondary-data method and a traceable evidence chain. O2 was met by developing the frozen-SigLIP compatibility pipeline and comparing its 379,265-parameter head with non-learned, linear, recurrent, type-aware and attention-based controls, alongside FITB, calibration, ablation, repeated-seed and overfitting analyses. O3 was met within the prototype boundary by integrating abstract candidates, exact search, weather and clothing-range constraints, local wardrobe management and a tested fallback. The interface uses a short home-page route, soft colour themes and an explicit menswear–womenswear selector; these are implemented design features rather than measured usability outcomes. O4 is addressed through the validity and limitation analysis below, which bounds the findings by FITB coverage, sparse menswear evidence, source-domain difference and the absence of participant evaluation.

6.7 Validity and confidence in the findings

Construct validity is limited by the operational definition of compatibility. The model learned from observed Polyvore outfits and same-category replacements. These labels are reproducible and cannot capture every reason why a person might combine garments. FITB is closer to candidate completion than binary classification, yet it remains an offline choice among four supplied answers. The resulting evidence therefore concerns ranking under this dataset-derived definition of compatibility.

Internal validity was strengthened by common splits, frozen features, fixed comparison settings, validation-based checkpointing and paired analysis. The local overlap audit found zero shared outfit identifiers, while non-zero item identifier overlap means that strict product-level disjointness was not verified and some memorisation risk remains. The comparison architectures were adaptations, and implementation choices may favour some models. Three seeds reveal only part of the training variation and do not characterise all hyperparameter uncertainty. Bootstrap intervals describe uncertainty conditional on the retained samples and scoring procedure; they do not correct sampling bias in the original data (Efron and Tibshirani, 1994; Bouthillier et al., 2021).

External validity is more restricted. The feature-domain classifier separated Polyvore and Zara embeddings with AUC 1.000. This shows that the two sampled sources were readily distinguishable in the chosen representation. It does not quantify how much compatibility performance would change on new retail or user photographs. The four-position representation excludes dresses and other multi-position garments, and only 29.50% of official FITB questions were evaluable. Menswear evidence is particularly sparse. The scope of the findings is limited to similar processed data and supported garment structures.

Ecological validity is also untested because no participant study was conducted. The application demonstrates that its components can be integrated and that tested actions function; it does not demonstrate usefulness in daily life. This distinction follows the broader finding that recommender evaluation requires metrics matched to the intended task and that offline accuracy alone is incomplete (Herlocker et al., 2004; Deldjoo et al., 2023).

6.8 Contribution, benefit and responsible use

The principal contribution is not a new foundation model. It is an auditable combination of reused visual representations, a small relational compatibility head, con-

trolled evaluation and constrained abstract search. The implementation shows how a resource-limited project can investigate compatibility without training a large image encoder end to end. Separating learned scoring from feasibility rules also makes it possible to identify whether a failure arose from model ranking, candidate availability or a contextual constraint.

Product-independent output provides a practical design benefit. A recommendation can describe a broad type and colour without suggesting that one brand contains the only suitable product. The selected garment remains visible and fixed, while abstract icons distinguish generated suggestions from garments saved by the user. Local wardrobe storage and the absence of automatic gender inference are consistent with the prototype’s privacy boundary. These are implemented properties, not measured user benefits.

A later version could link abstract recommendations to optional product matches from participating retailers or independent designers. This prospective route would require verified permissions, current product data and separate commercial evaluation.

6.9 Implications for applied AI practice

The results suggest that model scale should follow the problem rather than be treated as the main indication of technical quality. The proposed head trained faster than the controlled recurrent model and used a similar or smaller number of trainable parameters than the neural alternatives. Its stronger result came from an architecture that matched the relational task, especially the explicit pairwise summary. Once a transferable representation is available, a focused downstream component can be sufficient for a bounded application.

The development process also shows the value of treating data transformation as part of the model. The four-position mapping, construction of negatives, FITB eligibility rules and clothing-range labels determine which relationships the model can learn and which cases can be evaluated. These choices are not neutral preparation steps. For example, the exclusion of multi-position garments directly caused part of the FITB coverage loss, while sparse menswear-labelled data restricted subgroup conclusions. An applied system should version its data mapping and report coverage together with performance, rather than presenting only the final checkpoint.

The same principle applies to candidate generation. The learned score did not produce useful recommendations until possible garment states were represented, filtered and searched. The exact-search experiment showed that an apparently efficient quota

could change the result before the model had an opportunity to score most combinations. In a larger system, retrieval quality would need its own acceptance criteria. Improving the ranker while leaving an unmeasured retrieval bottleneck could produce no visible improvement for the user.

Finally, saved configurations, fixed seeds, hashes, separate evaluation outputs, explicit constraints, failure-aware loading and a rule fallback make the path from evidence to interface more inspectable. The neural representation remains only partly interpretable; these controls reduce ambiguity about what was trained, selected and deployed. For an MSc Applied AI project, this system-level traceability is part of technical competence because it connects an experimental result to reproducible software behaviour.

6.10 Prioritised future work

Future work should address evidence gaps before adding complexity. The first priority is licensed data expansion. A new dataset should include broader menswear coverage, documented provenance and explicit structures for dresses and other one-piece garments. This would require data mapping, new position occupancy rules, regenerated frozen embeddings, retraining and repetition of the AUC, FITB, ablation and subgroup evaluation. Dresses should not simply be relabeled as inner tops because that would incorrectly permit a separate bottom.

The second priority is target-domain validation. A modest, legally reusable set of complete outfits from a contemporary target domain should be labelled under a documented protocol. It could test whether compatibility ordering transfers beyond Polyvore and whether calibration remains valid. If performance falls, carefully bounded encoder adaptation or domain-robust training could be compared with the frozen baseline. Reliable data and permission review are the main requirements, alongside additional computing time.

The third priority is participant evaluation after ethics approval. A study could compare recommendations from the learned model, the rule fallback and a controlled baseline while measuring perceived usefulness, choice diversity and trust. Recruitment, informed consent, withdrawal, secure data handling and a defined analysis plan would be required.

The fourth priority is candidate-search scaling. Exact enumeration should remain the reference while the current candidate space is bounded. If a larger catalogue makes latency unacceptable, structured pruning or approximate search should be evaluated

against exact top- k recall, score loss and runtime. This would preserve a measurable accuracy–efficiency trade-off.

Chapter 7: Conclusion

This dissertation investigated whether a small project-specific compatibility head on frozen vision–language embeddings, combined with constrained search over abstract garment states, could support practical and auditable complete-outfit recommendation. The result is a working prototype and a controlled body of offline evidence.

The primary research question can be answered positively within the evaluated boundary. The head showed stronger held-out discrimination than the non-learned and controlled learned alternatives, while corrected FITB evaluation supported completion ranking on the retained subset. Explicit pairwise comparison had the clearest architectural support: removing it reduced mean FITB accuracy by 10.28 percentage points. The effect of clothing-position embeddings remained inconclusive.

Held-out test performance closely matched validation performance, supporting the consistency of checkpoint selection within the processed Polyvore split. The larger train–test gap showed overfitting, and the domain diagnostic warned against assuming unchanged performance on contemporary catalogue images. These results justify the use of early stopping and several aligned evaluation measures rather than selection by one favourable score.

The system contribution extends beyond classification. Product-independent states describe broad garment types and colours without making one retailer the recommendation source. Exact constrained search improved completeness under the deployed model objective for the bounded candidate space. Weather, clothing-range, fixed-item and position rules remained explicit, while local wardrobe management and a deterministic fallback separated learned scoring from operational control.

The findings remain limited by 29.50% FITB coverage, sparse menswear evidence, the Polyvore–Zara domain distinction, excluded one-piece garments and the absence of participant evaluation. The principal contribution is an evidence-backed, auditable integration of a frozen representation, a focused relational head, controlled comparisons, exact search and explicit constraints. Future work should prioritise licensed contemporary evaluation data with broader garment coverage, followed by a participant study if ethical approval is obtained.

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Chapter A: Evidence of Required Ethics Training

The following records were supplied by the student as evidence of completing the required training. The two-page Epigeum certificate records completion of *Research Integrity: Core* with an overall score of 97% on 11 May 2026. The subsequent University of Warwick Moodle emails record completion of *Information Security Smart* and award of the *WMG Student Ethics Training* badge.

Certificate

Number: 3867390383

This is to certify that

Junyuan Yao
of University of Warwick

Successfully completed the course
Research Integrity: Core

Module Title

Responsible research practice
Irresponsible research practice
Planning your research
Managing and recording your research
Data selection, analysis and presentation

Score (%)

90%
100%
100%
90%
100%

as part of the Epigeum Online Course System with a score of 97%.

Completion Date: 11 May 2026

Copyright Sage Publications 2023

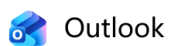
Figure A.1: Research Integrity: Core certificate, page 1 of 2



Figure A.2: Research Integrity: Core certificate, page 2 of 2

2026/5/12 00:23

收件箱 - YAO, CLIFF (PGT) - Outlook



Course completed

发件人 Do not reply to this email (via Moodle@Warwick) <no-reply-moodle@warwick.ac.uk>

日期 周二 2026/5/12 7:00

收件人 YAO, CLIFF (PGT) <Junyuan.Yao@warwick.ac.uk>

Congratulations!

You have completed the course [Information Security Smart](#).

Are you reading this in an email? [Download the mobile app and receive notifications on your mobile device](#)

Figure A.3: University of Warwick Moodle confirmation of Information Security Smart completion

2026/5/12 00:24

收件箱 - YAO, CLIFF (PGT) - Outlook



Congratulations! You just earned a badge!

发件人 moodle.warwick (via Moodle@Warwick) <no-reply-moodle@warwick.ac.uk>

日期 周五 2026/5/8 20:09

收件人 YAO, CLIFF (PGT) <Junyuan.Yao@warwick.ac.uk>

 1 个附件(352 KB)

WMG_Student_Ethics_Training.png;

You have been awarded the badge "WMG Student Ethics Training"!

More information about this badge can be found on the [WMG Student Ethics Training](#) badge information page.

You can manage and download the badge from your [Manage badges](#) page.

Are you reading this in an email? [Download the mobile app and receive notifications on](#)

Figure A.4: University of Warwick Moodle confirmation of the WMG Student Ethics Training badge

Chapter B: Completed Supervisor Delegated Approval Record

The following five-page Qualtrics response summary is the complete Supervisor Delegated Approval record supplied by the student. It identifies Junyuan Yao, student ID 5736642, the MSc Applied Artificial Intelligence course and the supervisor, Giovanni Strano. The record gives student ethics-form version 1 and confirms that ethical approval was waived because the research uses publicly available secondary data only. It records the earlier concise project title *AI Outfit Recommendation System Using Computer Vision*.

Response Summary:

Supervisor Delegated Approval (SDA) Form for WMG Taught Student Projects

This form is NOT for supervisors of cyber security students

Important information:

- This form is for taught student projects ONLY. EngD/PhD/Postgraduate Research students must receive ethical approval from BSREC
- Approval can be delegated to taught student projects that are LOW RISK (click [here](#) for information)
- On submission of this form, a notification email will be sent by Qualtrics. The email confirmation of approval/waiver must be included in the student's project submission

What is this form about?

All supervisors/coaches must grant or waive ethical approval for their student's research project BEFORE the student carries out data collection. There are four possible outcomes a supervisor can delegate to a student's ethics form: approve, waive, defer, and not approve.

This form is a legally binding agreement, therefore, when you sign and submit the form you are acknowledging your role as a supervisor in advising the student to maintain high professional standards of ethical conduct as set down in UK legislation. This includes maintaining the ethical principles and guidance provided by WMG and the University of Warwick.

When is the deadline to submit this form?

Supervisors/coaches must submit this form as soon as possible to enable the student to complete their research project and dissertation with sufficient time prior to the deadline. If it is not possible to grant or waive ethical approval, then it is important the supervisor/coach:

- Provides feedback to the student so necessary changes can be made to the ethics form to delegate approval/waiver
- Contacts the WMG Projects Team to inform them of any delays to the student's project completion
- Completes this form selecting the option "Ethical approval is not granted" and provides an explanation for this selection **OR** "Ethical approval requires deferral" and provides an explanation for this selection

Who needs to fill in this form?

The supervisor/coach is responsible for completing this form for each of their project students. A student will NOT be able to complete this form as it is for *supervisor* delegated approval of student research.

What information will be asked?

The form seeks information about the supervisor/coach, the project student, the student ethics form, and confirmations related to ethical approval.

To complete this form, please have the following information ready:

- Student's ID number and email address
- A PDF of the student's ethics form

To generate a PDF of the student's ethics form, print the email from Qualtrics containing the

responses to the student ethics form and 'save as PDF'.

How many times can a supervisor submit this form?

This form can be submitted as many times as needed. Students must include the email confirmation of ethical approval/waiver in the Appendices of their dissertation. *Note this email confirmation of ethical approval/waiver will be sent to the supervisor and student after successfully submitting this form.*

How much time will it take to complete this form?

It is estimated that the form will take no more than 15 minutes to complete, including uploading a PDF of the student ethics form.

How will the data collected in this form be used?

The data from this form will be shared with the Projects Team, Course Leaders, Heads of Group, Programme Administrators, and Ethics Committees.

Will my data be safe?

Data will be securely stored on Warwick platforms.

What if I and/or my student has not received the email notification for ethical approval/waiver following submission of this form?

Please wait at least five minutes for the email notification to appear. You may need to check your Spam/Junk folder if the email does not appear in your inbox. A supervisor/coach is responsible for ensuring the student receives an email confirmation of ethical approval/waiver. This means a supervisor/coach may need to forward the email notification they have received from Qualtrics directly to their student(s).

Typically, if an email notification from Qualtrics is not received, it is due to an incorrect email address being provided when completing the form. You may need to resubmit the form if there is an error. If after resubmitting the form and correcting email addresses a notification email from Qualtrics does not appear, please contact the Projects Team.

What if I have a question about this form?

- If you have a question about the project requirements of the student's course, contact the Course Leader
- If you have a question about research ethics, contact Helen.Ascroft@warwick.ac.uk

Q1.2. What programme is your project student(s) enrolled on?

- Full-time Postgraduate Taught programme (e.g. FT MSc)

Q1.3. Are you delegating an ethics outcome to a group student research project, such as an Industry Impact Project?

- No

Q1.4. Is this your first year supervising a taught student project at WMG?

- No

Section 1. Student Information for Individual Research Project

All questions must be answered with the correct information to ensure the student receives an email notification of ethical approval/waiver.

Q2.2. What is the student's name? *Please check this is filled in correctly by copying the information from the student's Tabula profile.*

Junyuan Yao

Q2.3. What is the student's Warwick ID number? *Please check this is filled in correctly by copying the information from the student's Tabula profile.*

5736642

Q2.4. What is the student's Warwick email address? *Please check this is filled in correctly by copying the information from the student's Tabula profile.*

Junyuan.Yao@Warwick.ac.uk

Q2.5. What is the student's project title? *This information can be copied from the student's ethics form.*

AI Outfit Recommendation System Using Computer Vision

Q2.6. What full-time master's course is the student enrolled on? *If the course is not listed, return to the first page of this form to select the correct programme.*

- MSc Applied Artificial Intelligence

Section 2. Supervisor/Coach Information

All questions must be answered with the correct information to ensure you receive an email notification of ethical approval/waiver. You can locate information about your staff ID and Warwick email address on your Tabula profile.

Q4.2. What is your name?

Giovanni Strano

Q4.3. What is your university ID number? *This can be found on your Tabula profile.*

1770819

Q4.4. What is your Warwick email address? *Please check this is filled in correctly so you can receive an email notification following submission of this form.*

g.strano.1@warwick.ac.uk

Section 3. Confirmations of Research Ethics and Upload of Student Ethics Form

All statements must be responded to in order to complete this form.

Q5.2. I have checked the student ethics form and can confirm that:

- 1. I am satisfied that the student has completed ALL mandatory training as required by the University: **WMG Ethics Training, Epigeum, and Information Security Smart**
- 2. I have discussed the importance of ethical approval with the student(s) and am satisfied that they understand the importance of maintaining high ethical standards when conducting research.
- 3 - FT MSc. I have discussed with the student(s) that data collection must take place whilst they are resident in the UK.
- 4. I have discussed with the student(s) that one of the primary conditions of being granted ethical approval/waiver is the requirement to work with their supervisor/coach to develop their research methods, approach and tools.

Q5.3. Please indicate if any of the following cases has been mentioned in regard to the student's ethics form:

- None of the above

Q6.1. Please upload a PDF record of the student ethics form that you are granting/waiving ethical approval for. *Note if the student has created multiple versions of the ethics form, it is imperative that only the approved version is uploaded here.*
[\[Click here\]](#)

Q6.2. Please enter the version number of the student ethics form that you are approving. *This is to ensure the correct ethics form is being approved in case the student has submitted multiple ethics forms.*

1

Q6.3. By submitting this form, I confirm:

- Ethical approval has been **waived** for this research, which uses publicly available secondary data **only**.

Embedded Data:

N/A

Chapter C: Supplementary Experimental Results

C.1 Clothing-range coverage in the retained FITB subset

The retained official FITB subset contained 46 menswear-labelled questions and 4,422 womenswear-labelled questions. Descriptive accuracy was 0.4348 for the menswear subset and 0.6043 for the womenswear subset. The menswear result is not treated as a reliable group comparison because its sample is substantially smaller.

Chapter D: Data Sources and Reproducibility

This appendix provides direct links to the external resources used in the project. The research method, dataset preparation and evaluation procedure are described in Chapters 3 and 4.

- **Polyvore Outfits disjoint dataset:** authors' code and dataset repository (Vasil-eva et al., 2018b). This was the main source for compatibility training and evaluation.
- **Maryland Polyvore metadata:** dataset repository. This supplied supporting product-category metadata.
- **Zara Products:** Kaggle dataset page (Parreño Lara, 2024). This was used for recognition preparation and the feature-domain comparison reported in the dissertation.
- **SigLIP:** model card and checkpoint page (Google, 2024). The encoder was used in frozen form.
- **Open-Meteo:** forecast API documentation (Open-Meteo, 2026). This supplied the weather context used by the prototype.

The project repository contains the code, tests and machine-readable result files used to record the experiments. Key model settings, sample sizes, confidence-interval procedures and limitations are reported in the main chapters so that the findings can be interpreted without relying on this appendix.

Chapter E: Supplementary Software-Design Artefacts

This appendix documents the implemented prototype at a level below the methodology diagrams. The views use UML conventions where they match the function-oriented code and represent persistent entities as a logical local file schema.

E.1 Use cases

Figure E.1 records the actions available to the single local wardrobe user and the external weather dependency. Saving and wardrobe management remain local actions; the weather service receives a city query but does not receive wardrobe images.

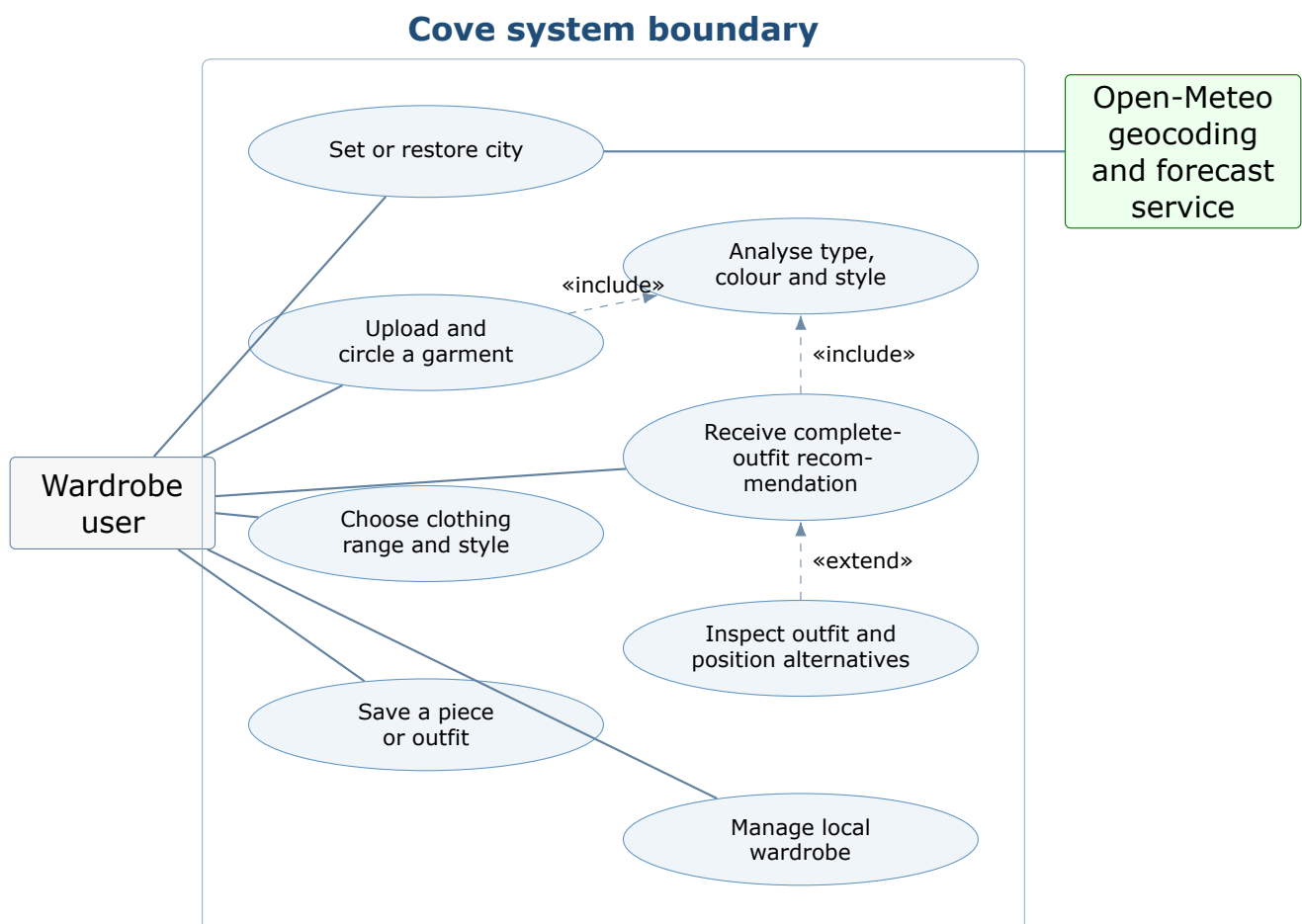


Figure E.1: UML use-case view of the implemented Cove prototype

E.2 Overall software structure

The application is predominantly function-oriented. Figure E.2 therefore uses a UML-style component view to show the implemented source modules, external service and stored artefacts.

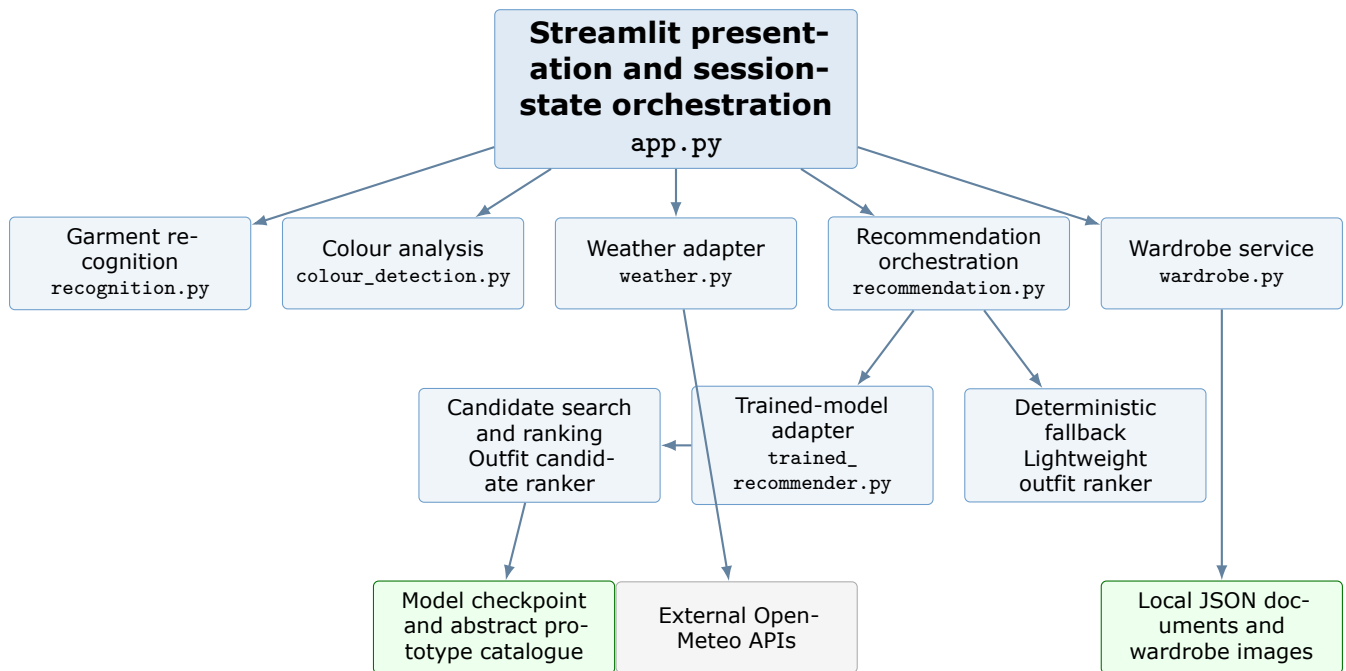


Figure E.2: Overall UML-style component view of the Cove software

E.3 Recommendation-module classes and dependencies

Figure E.3 focuses on the classes that are present in the recommendation path and includes the function-based adapter and facade on which they depend.

E.4 Logical entity relationships

Figure E.4 expresses the multiplicities in the local wardrobe schema. It is an entity-relationship view of serialised application state, not evidence of a relational database implementation.

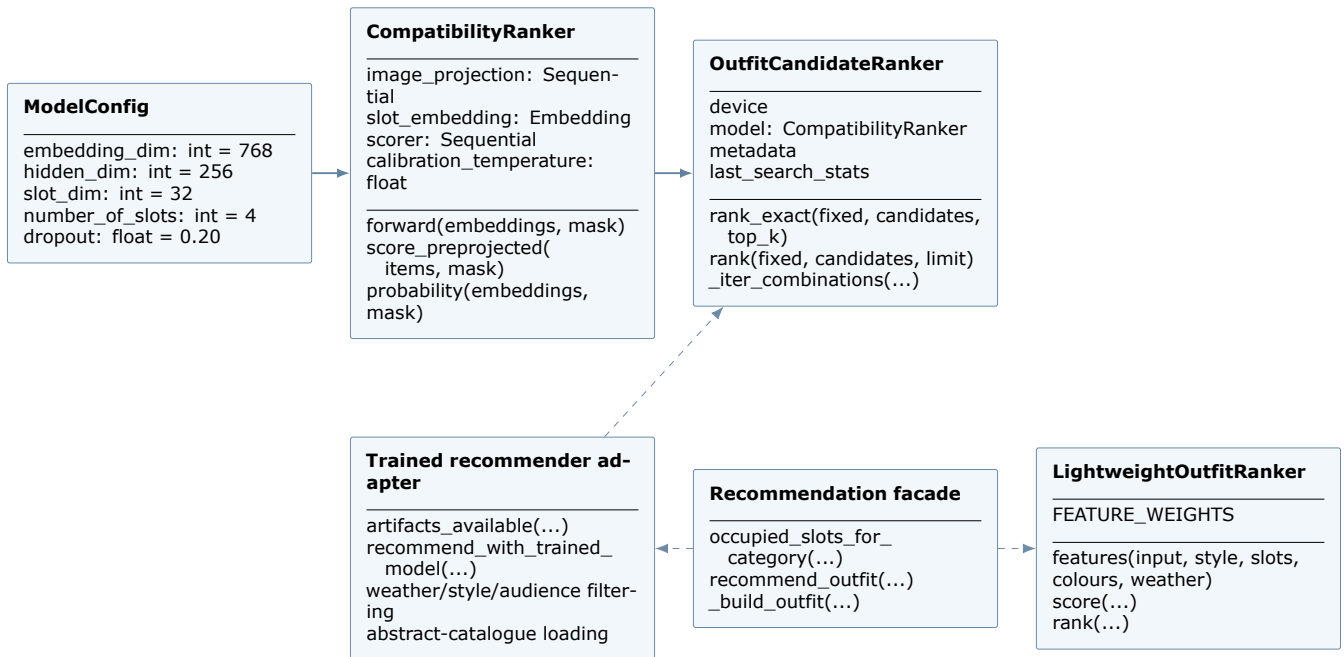


Figure E.3: Module-level UML class and dependency diagram for recommendation

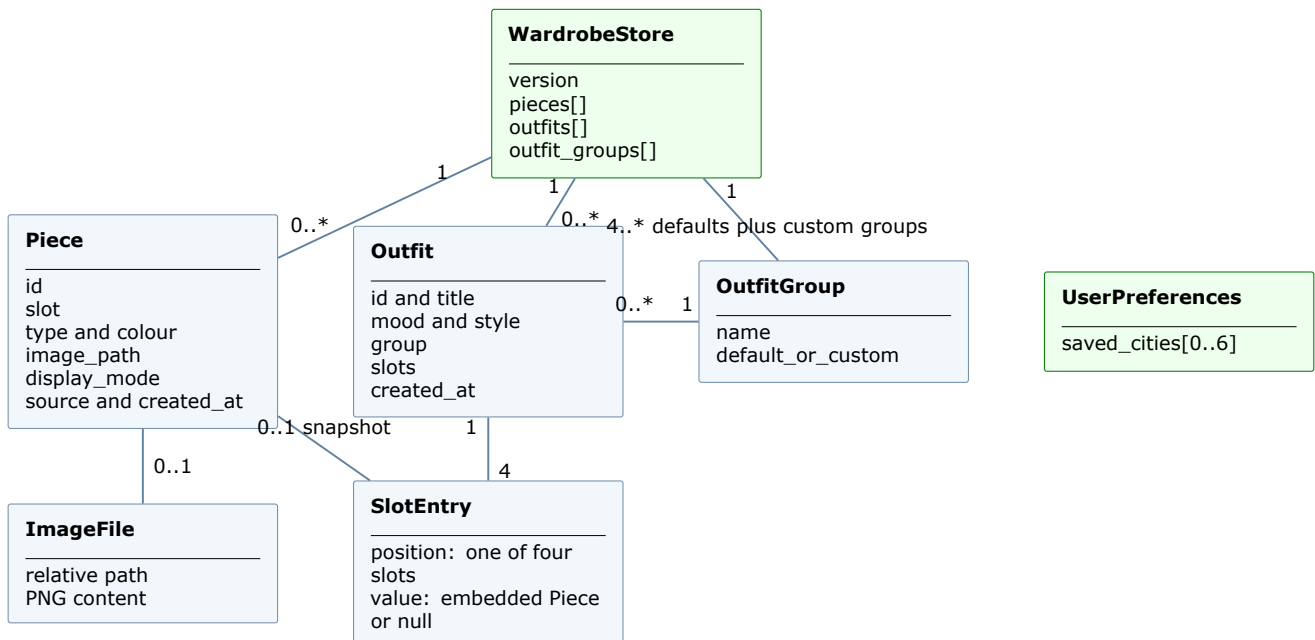


Figure E.4: Entity-relationship view of the logical file schema; these entities are serialised to JSON rather than relational tables

E.5 User stories and acceptance criteria

Table E.1 converts the implemented interaction paths into testable user stories. Priorities distinguish the core recommendation journey from supporting convenience features.

Table E.1: Implemented user stories and acceptance criteria

ID	Feature	User story	Acceptance criteria	Priority
US-01	Location	As a user, city weather should inform the outfit context.	A valid city returns current conditions; an invalid or unavailable request produces a bounded error rather than stopping the application.	Must
US-02	Saved cities	As a user, frequently used cities should be available without retyping.	A city can be saved or removed; no more than six are retained; selecting one refreshes the forecast.	Should
US-03	Image selection	As a user, a garment should be selectable from a photograph.	The user can upload an image, draw a region and repeat the selection before analysis.	Must
US-04	Recognition	As a user, the selected garment should be interpreted before recommendation.	The application returns a supported broad category, colour, style cues and an image embedding, with an editable fallback when recognition is uncertain.	Must
US-05	Context control	As a user, clothing range and style should remain under explicit control.	Menswear or womenswear and an available style can be selected; changing either regenerates the recommendation without inferring gender from the image.	Must
US-06	Outfit completion	As a user, the selected garment should remain fixed while the outfit is completed.	The result covers all feasible supported positions, respects occupied slots and weather constraints, and uses product-independent types and colours.	Must
US-07	Alternatives	As a user, more than one acceptable solution should be inspectable.	The interface provides a primary outfit, a materially different alternative and model-ranked position replacements where feasible.	Must
US-08	Operational fallback	As a user, recommendation should remain available if model artefacts cannot load.	The system records the loading reason and returns a deterministic rule-ranked result without presenting it as a learned score.	Must
US-09	Save result	As a user, either the supplied garment or a complete recommendation should be saved.	Saving preserves the user image where available and represents abstract recommended garments as coloured icons.	Should
US-10	Outfit editing	As a user, saved outfits should be completed and revised later.	Each of four equal positions supports add, replace or confirmed removal; an outfit can be renamed and assigned to a group.	Should
US-11	Piece library	As a user, individual garments should be reusable across outfits.	Pieces can be filtered by four positions, renamed, reused in multiple outfits and deleted with confirmation.	Should
US-12	Local persistence	As a user, wardrobe information should remain private on the device.	Wardrobe JSON, preferences and images are stored locally and are not incorporated into model training.	Must